

SEASONAL LANDSLIDE INVENTORY FOR NER-2024



North Eastern Regional Node for Disaster Risk Reduction (NER-DRR)
NORTH EASTERN SPACE APPLICATIONS CENTRE
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12	Abstract	The study is an ongoing process under NERDRR programme of NESAC, Umiam. It is basically a compendium of reported landslide incidents compiled from print and online media reports. A total of 1137 landslide incidents have been documented for the year 2024 from the different states of the NER, mainly triggered by rainfall. Out of these landslides 914 landslides was reported by Tripura SDMA for August and September, 2024. Further, 98 casualties have been reported out of which 1 is from Arunachal Pradesh, 10 from Assam, 5 from Manipur, 13 from Meghalaya, 38 Mizoram, 7 Nagaland, 13 from Sikkim and 11 are from Tripura. From various locations collapse of road-sections along National/State Highways as well as district roads are reported and are also documented. This report also contains information on 24 hrs total rainfall prior to each event collected from GPM Mission. Wherever possible event locations are verified in terms of precise latitude and longitude, either through field/UAV survey, post-dated satellite imagery and/or media photographs. Accordingly, codification scheme has been assigned. Landslide dimension vis-a-vis spatial resolution of the satellite image, availability of cloud free image, exposure condition of the landslide with respect to hill slope, clearance of slide debris especially along roadside immediately after an event are some of the factors that pose limitations for validation of each and every reported event location through (freely available cloud free optical) satellite data. It also needs to be mentioned that the report is not inclusive of all landslide events that have occurred during the year 2024. The systematic inventory thus compiled will help in assessing the risk in terms of yearly loss of life and property, disruption of transportation network, frequency and magnitude of landslide incidents in an area and NER as a whole. It will also help in carrying out studies related to landslides in NER.		
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PREFACE

North Eastern Regional node for Disaster Risk Reduction (NER-DRR) has been set up at NESAC, Umiam in the year 2011 to provide scientific solutions towards disaster management to the entire North Eastern Region (NER) of India. Eight states of NER of India, namely, Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim and Tripura are susceptible to landslides except the plain areas of Assam, Manipur and Tripura. Physiographically NER is apart of Extra-Peninsula of India and geologically recent in origin, having exposed rocks from Archaean to Quaternary ages as well as tectonically active. These diverse setup and high mountain ranges make the terrain of NER susceptible to landslides. While, intense and/or prolonged monsoonal rainfall and seismic activities are the immediate trigger; anthropogenic factors have a role in accelerating the issue too. Hence, the need for systematic cataloguing and preparation of a date and time tagged landslide inventory databank is realized highlighting the landslide locations with respect to nearby settlements and transportation network immediately after the event is reported. Accordingly, vector layer is updated for each and every reported event immediately at <https://www.nerdrr.gov.in> while report is finalized after proper validation of the reported events post availability of cloude free satellite image.

It is expected that the systematic inventory thus compiled will help in visualizing seasonal landslide events in spatial context and assessing the risk in terms of yearly loss of life and property, disruption of transportation network, frequency and magnitude of landslide incidents in an area and disaster resilient developmental planning.



DIRECTOR, NESAC

Contents

Preface	v
1. Introduction	1
2. Methodology and Data used	2
2.1 Landslide point and polygon extraction	2
2.2 Data Source	3
2.3 Automatic Extraction of News from Online Media	3
2.4 Satellite Data	4
2.5 Rainfall Data	4
2.6 Automatic Rainfall Data Extraction	5
3. Details of the landslide inventory prepared for 2024	5
4. Landslide Inventory in different parts of the NER	10
4.1 Highlights of Landslide incidents in Arunachal Pradesh	11
4.2 Highlights of Landslide incidents in Assam	12
4.3 Highlights of Landslide incidents in Manipur	12
4.4 Highlights of Landslide incidents in Meghalaya	13
4.5 Highlights of Landslide incidents in Mizoram	14
4.6 Highlights of Landslide incidents in Nagaland	16
4.7 Highlights of Landslide incidents in Sikkim	16
4.8 Highlights of Landslide incidents in Tripura	17
5 References	19
6. Annexures	21
Annexure-I: Landslide Incident Reports	21
Annexure-II: Landslide Verification on Satellite Imagery and Google Earth Imagery	38
Annexure-III: Newspaper (Clippings & Digital Archive)/ Online Media	40
Annexure-IV: Field Survey Photographs	42
Annexure-V: Inventory Attributes	44

List of Figures

Figure 1	Flowchart for landslide polygon extraction	2
Figure 2	Flowchart for automatic extraction of rainfall	5
Figure 3	Landslides and casualties of NER, India in 2024	9
Figure 4	Monthwise landslides in NER, India (2024)	10

List of Tables

Table 1	Specifications of the stellite data used.....	4
Table 2	Specifications of the GPM data used.....	5
Table 3	Statewise list of newspapers reporting landslide incidents in NER.....	6
Table 4	Landslide statistics of NER, India in 2024	9
Table 5	Landslide casualties of NER, India in 2024	10

List of Maps

Map 1	Landslide incidents in the NER, 2024	8
Map 2	Landslide incidents in Arunachal Pradesh, 2024.....	11
Map 3	Landslide incidents in Assam, 2024	12
Map 4	Landslide incidents in Manipur, 2024	13
Map 5	Landslide incidents in Meghalaya, 2024	14
Map 6	Landslide incidents in Mizoram, 2024.....	15
Map 7	Landslide incidents in Nagaland, 2024.....	16
Map 8	Landslide incidents in Sikkim, 2024.....	17
Map 9	Landslide incidents in Tripura, 2024	18

1. Introduction

Landslide is described as downslope movement of soil, rock, and slope forming materials characterized by either very slow to sudden and rapid movement (Varnes, 1978). Landslides occur throughout the world in hilly terrain and under all climatic conditions, cost billions in monetary losses, and are responsible for thousands of deaths and injuries each year. Often, they cause long-term economic disruption, population displacement, and negative effects on the natural environment (Highland and Bobrowsky. 2008). According to the official data provided by the United Nations International Strategy for Disaster Reduction (UN/ISDR) and the Centre for Research on the Epidemiology of Disasters (CRED) in 2006, landslides ranked as the third most deadly natural disaster (Vordzorgbed 2022). Landslides account for approximately 17% of all fatalities resulting from natural hazards worldwide (Raja et al. 2017). Recent global landslide assessment studies (Petley et al. 2005; Nadim and Kjekstad 2009; OFDA/CRED 2010) reveal that the countries with the highest risk to landslides are mostly in the developing world. According to Geological Survey of India (GSI) reports, 427 k sq. km area accounting 12.6% of the Indian landmass is landslide prone and this contributes approximately 8% of global landslide fatalities. Among the vulnerable areas, 0.18 million sq. km lie in the North Eastern Himalaya, encompassing Darjeeling and Sikkim Himalaya, while 0.14 million sq. km are found in the North Western Himalaya, including Uttarakhand, Himachal Pradesh, and Jammu & Kashmir. Additionally, 0.09 million sq. km of susceptible land are situated in the Western Ghats and Konkan hills, spanning Tamil Nadu, Kerala, Karnataka, Goa, and Maharashtra (Jain et al. 2023). The remaining 0.01 million sq. km of risk-prone region exists in the Eastern Ghats of Araku in Andhra Pradesh. Landslides are classified into different types on the basis of the type of movement and the type of material involved. Landslides can be triggered by natural factors, such as heavy rainfall, snowmelt, earthquakes, volcanic eruptions, and erosion, as well as human activities, including deforestation, construction, mining, and improper land use (Van Westen 2000). The extent of damage caused by landslides can vary from minor disturbances to catastrophic events that lead to significant loss of life and destruction of property and infrastructure. Understanding the underlying causes and mechanisms of landslides is essential for effective hazard assessment, prevention, and mitigation efforts to minimize their impact on vulnerable regions and communities and systematic inventory of landslide events is the prerequisite for this.

Under the purview of North Eastern Regional Node for Disaster Risk Reduction (NERDRR), a need for immediate cataloguing of landslide incidences and creation of Geospatial database is realized highlighting the landslide location with respect to nearby settlements and transportation

network immediately after the event is reported. Landslide inventories are one of the key requirements for landslide studies (Van Westen et al. 2008).

However, it should be noted that this database does not encompass all the events that have occurred during the year 2024 pan NER. Also, the inventory database thus generated is not similar to the inventory database being generated under the National project ‘Satellite Integrated Landslide Assessment and Alert System (SILAAS)’ by National Remote Sensing Centre (NRSC), ISRO Hyderabad. This is because purpose of this project is to prepare a date & time tagged landslide inventory databank which is the primary requirement for assessing rainfall threshold triggering various types of landslides in diverse terrain of NER. This in turn will help in preparing an effective landslide alert system for the region. However, in most of the instances it is hard to get time tagged events.

2. Methodology and Data used

2.1 Landslide point and polygon extraction

To create the Seasonal Landslide Inventory for the monsoon period, an extensive review of relevant articles concerning landslide incidents from major Daily Newspapers in NER was conducted. The locations of all landslide incidents were extracted from these newspaper reports, and an initial database was prepared in the form of a Point Shape file. From all the incidents, those deemed significant enough to be visible on IRS LISS 4/Sentinel 2 imagery are sorted, and a 10 km buffer was established around them to resolve any geo-location uncertainties. Subsequently, using Pre and Post Event Satellite Imagery once available, visible landslide polygons within the 10 km buffer were identified and added to the inventory.

The flowchart for the methodology is displayed in Figure 1.

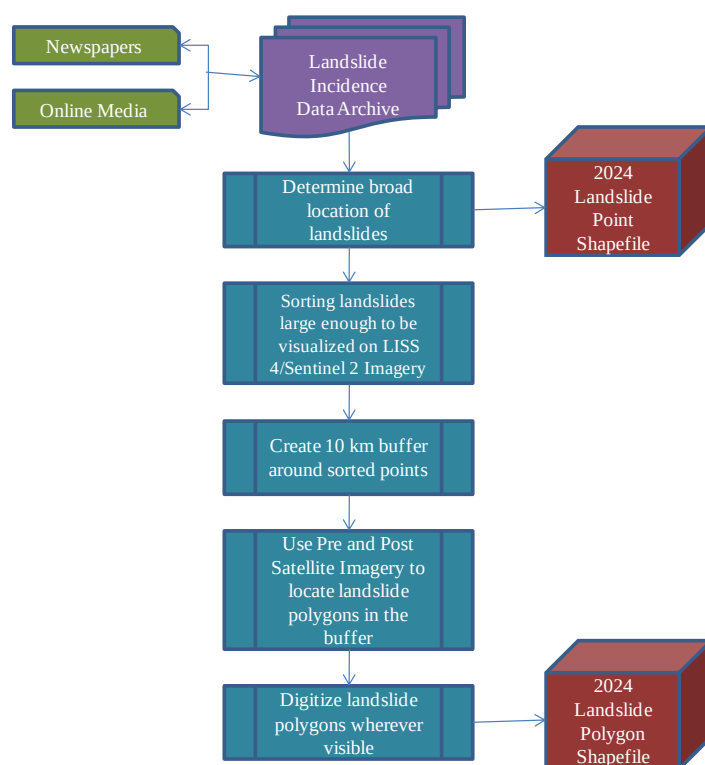


Figure 1 Flowchart of Landslide polygon extraction

2.2 Data Source

Unlike other disasters like earthquakes, so far, landslide incidents are not instrumentally recorded and reported anywhere immediately after the event. Hence, immediate cataloguing of landslide incidents has been done mainly from print (both local and national newspapers) and online/social media. Subsequently, each event location is verified in terms of precise latitude and longitude either through field survey or from post-dated satellite/UAV imagery as and when available. However, landslide dimension vis-a-vis spatial resolution of the satellite image, immediate availability of cloud free image, exposure condition of the landslide with respect to hill slope, clearance of slide debris especially roadside one immediately after an event are some of the factors that pose limitations for validation of each and every reported event location. Accordingly, codes are assigned for every event location. Further, due care is taken to minimise the confusion regarding date and time of occurrences and precise incident locations of reported events.

The data sources contributing to this inventory are categorized into three main groups:

1. Newspapers and online media as immediate information about a landslide event.
2. Satellite/UAV imagery, field photo and location for validation purpose.
3. Global Precipitation Measurement (GPM) mission for corresponding rainfall information.

2.3 Automatic Extraction of News from Online Media

To extract the information related to Landslide events relevant code has been developed by NERDRR IT team. Automatic news extraction script fetches news articles about natural disasters for defined Indian states using the GNews API. It retrieves news based on selected topics such as landslides, hailstorms, flash floods, earthquakes, forest fires, lightning, floods, heavy rainfall, thunderstorms, droughts, avalanches, and extremely heavy rainfall etc. for the states of NE: Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim, and Tripura. For the current purpose news related to landslides is only filtered. The date range is dynamically set to include only articles from the previous day. The fetched articles are filtered to ensure that the topic is mentioned in their descriptions and stored in a list. The data is then saved to a database, and the news entries are inserted into a table. The script checks each entry based on title and post time, to avoid duplication. After the database insertion and clean-up, changes are committed, and the connection is closed. This process automates the collection and storage of landslide hazard/disaster related news during monsoon season operationally from Mid May to end of October every year.

2.4 Satellite Data

Satellite imageries available post event or post monsoons are used to validate positional accuracy of the landslide locations and forextraction of affected area through polygon features. Freely available imageries from Resourcesat 2 (LISS IV MX) and Sentinel 2 Multispectral Instrument (MSI) is mainly used for extraction and updation of landslide features. The details of the sensor specifications are given in Table 1. Latest images from any of the satellite sensors freely available from Google earth platform are also used to validate the landslide inventory for gap areas (Annexure II).

Table 1 Specifications of the stellite data used

Satellite/Sensor	Spatial Resolution	Revist Time
LISS 4 Mx	5.8 m	5 days
Sentinel 2 (MSI)	10 m	5 days (Two Constellations)

A recently activated landslide area on hill slope appears bright due to exposure of fresh rock and soil and loss/removal of vegetation and other slope forming materials.

2.5 Rainfall Data

Rainfall data from NASA's Global Precipitation Mission (GPM) has been used, considering the seamless availability of rainfall data for the entire NER. The Global Precipitation Measurement (GPM) is an international mission directed by NASA and JAXA, focusing on the provision of advanced global observations of rain and snow. GPM Core Observatory serves as a reference standard combining data from a constellation of partner satellites to provide detailed precipitation measurements(gpm.nasa.gov).It involves a key product known as Integrated Multi-satellite Retrievals for GPM (IMERG). IMERG generates high-resolution estimates of precipitation that are derived through the combination of data from many satellites. Various IMERG products are available for different stages: Early, Late, and Final. The "Late Run" product, the GPM IMERG Late Precipitation L3 Half Hourly 0.1-degree x 0.1-degree V07 (GPM_3IMERGHHL), presents precipitation estimates nearly with a latency of 12 hours for implementing these timely applications. The GPM IMERG Late Run V07 product has a spatial resolution of 0.1° x 0.1°, or approximately 10 km x 10 km, and a temporal resolution of 30 minutes. The details of GPM product used is given in the Table 2 below.

Table 2 Specifications of the GPM data used

Plot Type	Variable	Unit	Source	Temporal Resolution	Spatial resolution
Time Series, Area Averaged	Multi-satellite precipitation estimates with climatological gauge calibration - Late Run (GPM_3IMERGHHL v07)	mm/hr	GPM	Half-Hourly	0.1 degree

2.6 Automatic Rainfall Data Extraction

From the list of landslides events, latitude, longitude and date of event are used to extract the corresponding rainfall data using automation pipeline created for this activity. This automation pipeline will first download the required rainfall data from specific sources (Currently GPM data supported) and pre-process them to get hourly precipitation and cumulative precipitation of each event. The flowchart for the methodology behind Automatic Rainfall Data Extraction is shown in Figure 2.

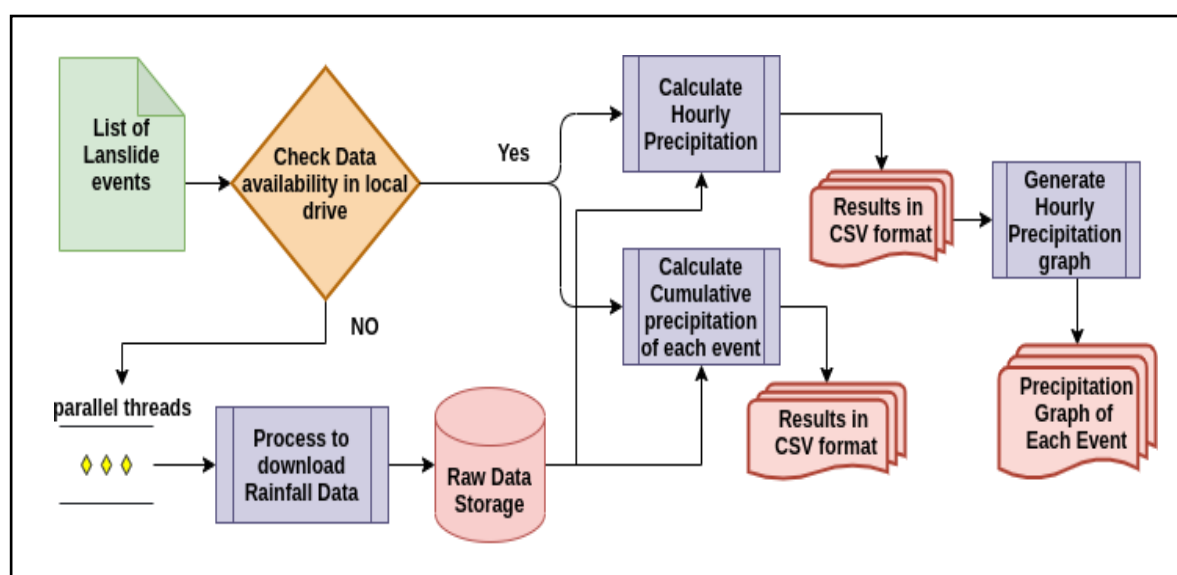


Figure 2 Flowchart for automatic extraction of rainfall

3. Details of the landslide inventory prepared for 2024

The various newspapers being used to gather information for the year 2024 is given in the Table3 and in Annexure III. Most of the leading daily newspapers distributed in the different states of NER were thoroughly reviewed in addition to online media which plays a significant role in reporting hazard and disaster incidents. Therefore, data collected from newspapers were supplemented with information acquired from online media too. In the context of this inventory; online media includes all the small local/regional news stations that only have an online

presence. With the ubiquitous accessibility of internet, often landslide incidents photographed/recorded by people are shared on different social media platforms. Certain community outreach groups also provide information about recent disasters on such sites. The army also provides updates on X (formerly twitter) whenever they are involved in rescue as well as in clearing landslide debris to facilitate transportation. Examples of the same used in this inventory are provided in the Annexure-III.

Table 3 Statewise list of newspapers reporting landslide incidents in NER

News Paper and Online Media	Arunachal Pradesh	Assam	Manipur	Meghalaya	Mizoram	Nagaland	Sikkim	Tripura
ABP Live							✓	
ANI News					✓			✓
Arunachal Observer	✓							
Arunachal Times	✓							
Arunachal24.in	✓							
Assam Tribune				✓			✓	
Borok Times								✓
Business Standard	✓							
CNBCTV			✓					
Deccan Herald				✓				
East Mojo			✓			✓	✓	
English Jagran		✓						
Guwahati Plus		✓						
Highland Post				✓				
Hindustan Times		✓	✓	✓				
Hindusthan Samachar			✓					
India Today			✓					
India Today NE	✓	✓		✓	✓	✓	✓	
IPR Nagaland						✓		
Ka Shlem		✓						
Morung Express						✓		
MSN	✓		✓	✓	✓	✓	✓	
Nagaland Post					✓	✓		
Nagaland Tribune						✓		
NDTV				✓				
Ne Now		✓						
NeNews				✓				
NeNow						✓		
News24Online					✓			
Northeast Live TV	✓			✓		✓	✓	
Northeast Today			✓				✓	
Pragativadi							✓	
Pragnews	✓							
PratidinTime	✓	✓		✓				
Scroll.In							✓	
Sentinel Assam	✓	✓		✓		✓		

Table Continued.....

News Paper and Online Media	Arunachal Pradesh	Assam	Manipur	Meghalaya	Mizoram	Nagaland	Sikkim	Tripura
Telegraph India	✓						✓	
The Print						✓		
The Sangai Express			✓					
The Shillong Times				✓				✓
The Week	✓							
Times of India				✓				✓
Tripura Info		✓						✓
Ukhrul Times			✓					
ZeeBusiness	✓							
Total Landslides	29	15	22	94	12	26	19	920*

**Out of these 914 landslides were reported by Tripura SDMA.*

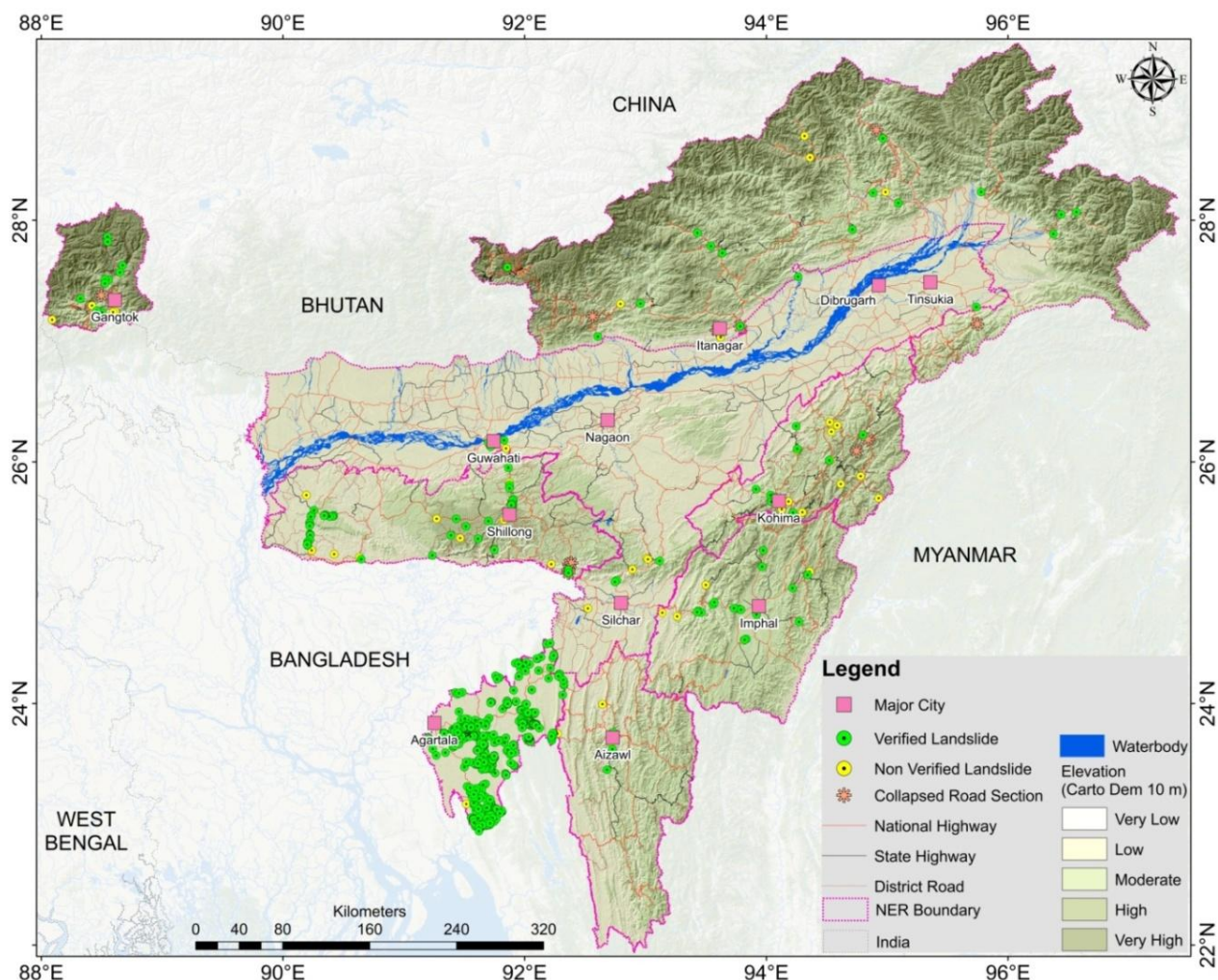
For the state of Tripura more than 2000 locations are reported as affected due to both flood and landslide, only confirmed landslide locations with information about date are included here.

#Newspaper on local languages were not referred as it can not be retrieved by the automatic code.

Across the different states of NER India, a total of 1137 landslide incidents resulting in 98 casualties have been reported. Among these incidents, 29 from Arunachal Pradesh, 15 from Assam, 22 from Manipur, 94 from Meghalaya, 12 from Mizoram, 26 from Nagaland, 19 from Sikkim and 920 from Tripura (Table 4,5, Map 1 & Figure 3). The month wise distribution of landslides is also shown in Figure 4. Moreover, the inventory also includes reports of road-section collapses along National/State Highways/other roads within the NER states. Base and other ancillary information used to correlate the inventory database are of 1:50,000 scale.

Cyclone Remal, which hit in late May 2024, had disastrous affect in Northeast India, in terms of landslides. Total 32 were people reported to lost their lives at different locations and incidents; major one being collapse of a stone quarry along National Highway 6 killing 28 people and even several remained missing. Fatal landslides were also reported from Hlumen, Salem, Falkawn, Aibawk, Lungsei, and Kelsih suffered in the vicinity of Aizawl.

Between August 18-25, 2024, the State of Tripura has experienced relentless rainfall that has triggered severe landslides and flooding events across multiple districts, leading to a humanitarian crisis of unprecedented scale. The total disaster has claimed 32 lives, injured three people, and left one person missing (Sphere India, 2024).



Map 1 Landslide Incidents in the NER, 2024

The different categories used post validation in the current inventory is given below:

- TYPE1(Represented by Green Colour)

These are the events, for which precise latitude/longitude information is available. Such events are generally validated from UAV/Satellite Imagery and/or Field Visits. Post-dated satellite imageries are used for validation purpose only where spatial resolution of the image, size and exposure condition of the landslide makes it possible.

- TYPE2(Represented by Yellow Colour)

These are the events, for which precise latitude/longitude information not available rather is referred with respect to nearby village/settlement or locality name by newspapers/media reports. In most of the case these slides are of smaller dimension and could not be visualized using freely available Resourcesat LISS IV MX and Sentinel 2 data.

- A 3rd category (TYPE 3) comprises areas where the impact of landslides on road sections along National Highways/State Highways/other roads are reported, resulting in disruptions of transportation. A different symbology is utilized to distinguish these locations. However,

validating these areas presents a challenge as road sections are typically restored or repaired promptly, leaving only a limited window for verification through field visits or satellite imagery. Additionally, the revisit status of a particular satellite further complicates the situation.

All the landslide locations also have standard attribute information, in terms of unique ID, location, date/time/type of events, casualty information, and 24hrs rainfall prior to the event etc. (Annexure V). Due care is been taken to validate all the event locations at the earliest.

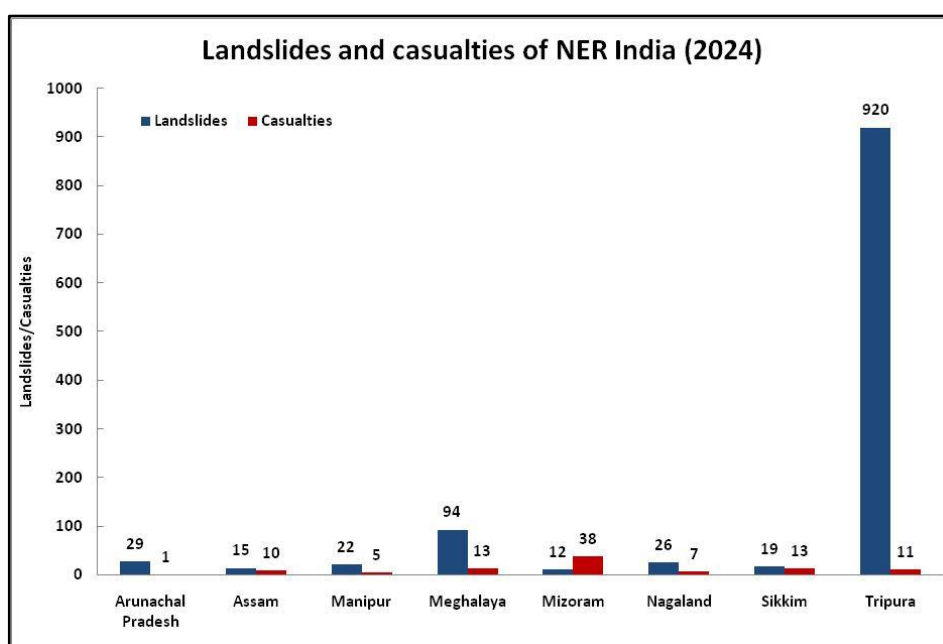


Figure 3 Landslides and casualties of NER, India in 2024

Table 4 Landslide statistics of NER, India in 2024

Months	Arunachal Pradesh	Assam	Manipur	Meghalaya	Mizoram	Nagaland	Sikkim	Tripura	Total Landslides
January	0	0	0	0	0	0	0	0	0
February	0	0	0	1	0	0	0	0	1
March	0	0	0	0	0	0	0	0	0
April	1	0	0	1	0	0	0	0	2
May	1	9	3	13	6	2	1	0	35
June	15	3	0	56	1	3	10	1	89
July	10	2	7	15	4	11	3	0	52
August	0	1	11	4	1	7	2	915	941
September	1	0	0	2	0	2	3	4	12
October	1	0	1	2	0	1	0	0	5
November	0	0	0	0	0	0	0	0	0
December	0	0	0	0	0	0	0	0	0
Total	29	15	22	94	12	26	19	920	1137

Table 5 Landslide casualties of NER, India in 2024

Months	Arunachal Pradesh	Assam	Manipur	Meghalaya	Mizoram	Nagaland	Sikkim	Tripura	Total Casualties
January	0	0	0	0	0	0	0	0	0
February	0	0	0	0	0	0	0	0	0
March	0	0	0	0	0	0	0	0	0
April	0	0	0	0	0	0	0	0	0
May	0	2	1	5	32	1	0	0	41
June	0	6	0	0	0	0	12	0	18
July	1	2	2	0	6	0	1	0	12
August	0	0	2	0	0	0	0	11	13
September	0	0	0	0	0	6	0	0	6
October	0	0	0	8	0	0	0	0	8
November	0	0	0	0	0	0	0	0	0
December	0	0	0	0	0	0	0	0	0
Total	1	10	5	13	38	7	13	11	98

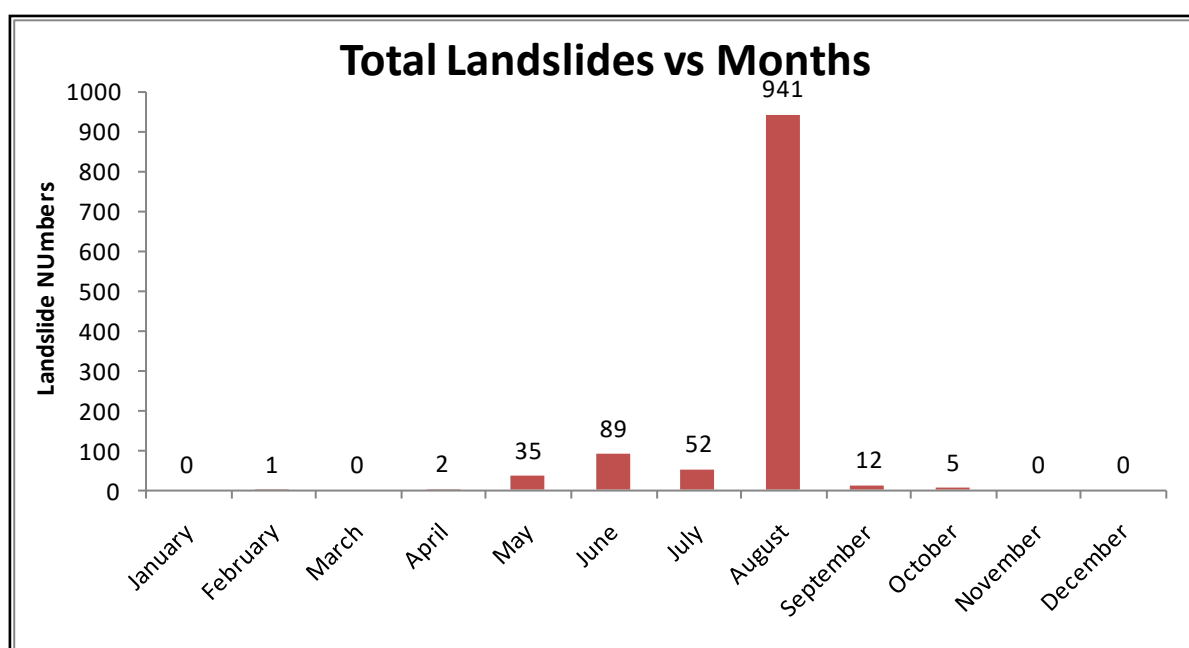


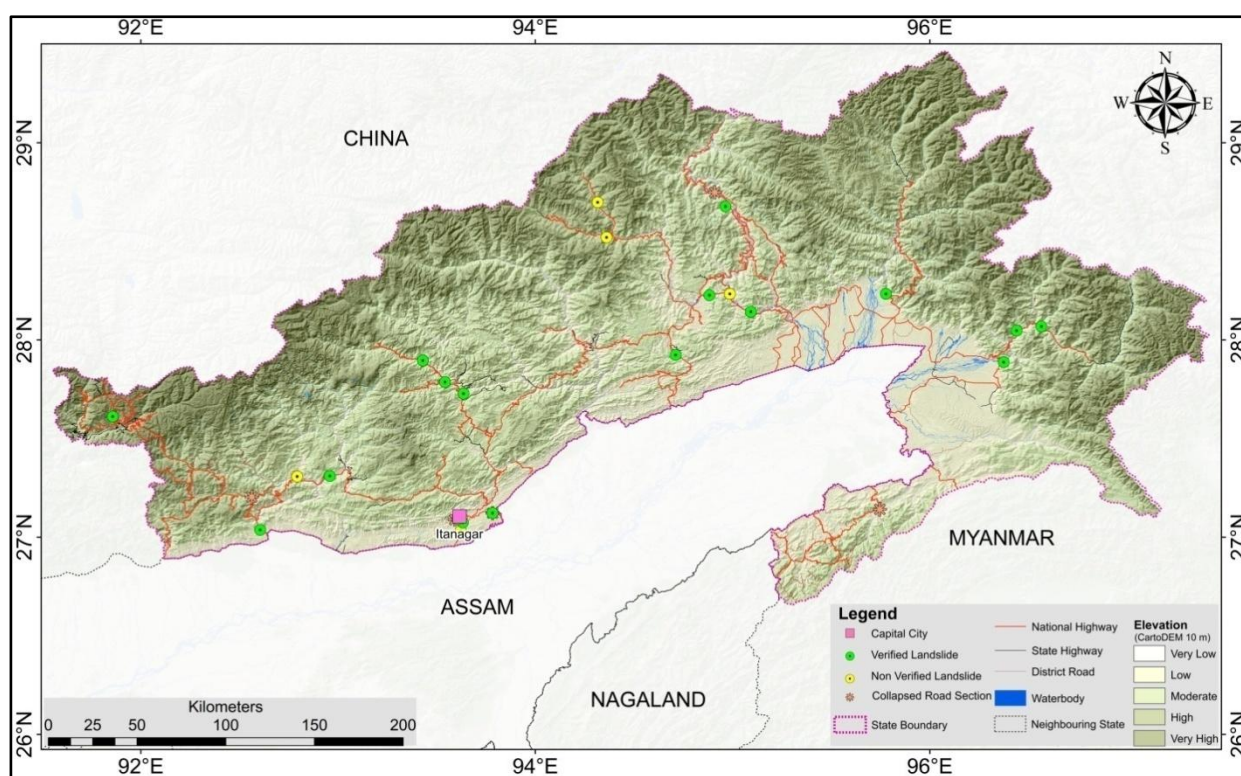
Figure 4 Monthwise landslides in NER, India (2024)

4. LANDSLIDE INVENTORY IN DIFFERENT PARTS OF THE NER INDIA

This section of the report depicts a concise overview of landslide incidents across the various states of the NER. Detailed information regarding each landslide incident is provided in the Annexure I.

4.1 Highlights of Landslide incidents in Arunachal Pradesh

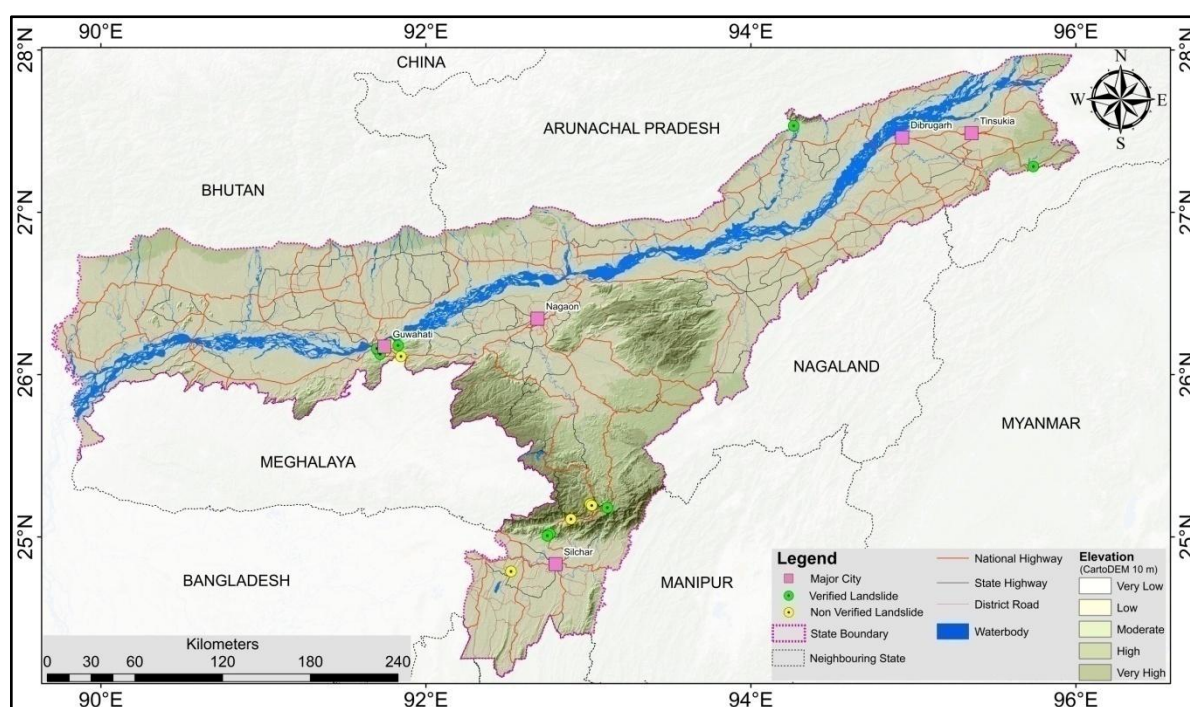
- A total of 29 landslide incidents recorded in Arunachal Pradesh.
- The first landslide incident for the year was recorded on 24th April 2024 and the latest event was recorded on 4th October 2024.
- There was 1 casualty occurred in Arunachal Pradesh in 2024.
- NH 313, between Hunli and Anini, NH-415 between Bandardeva and Nirjuli and NH-13 between Likabali-Basar-Daporijo Roads were some of the highways/road sections that were most affected in the state due to landslides.
- A landslide buried a school bus on the Ganga to Jullang road in Itanagar, Arunachal Pradesh, on June 29.
- The spatial distributions and details of the landslides are shown in Map 2 and Annexure I(I) respectively.



Map 2 Landslide incidents in Arunachal Pradesh, 2024

4.2 Highlights of Landslide incidents in Assam

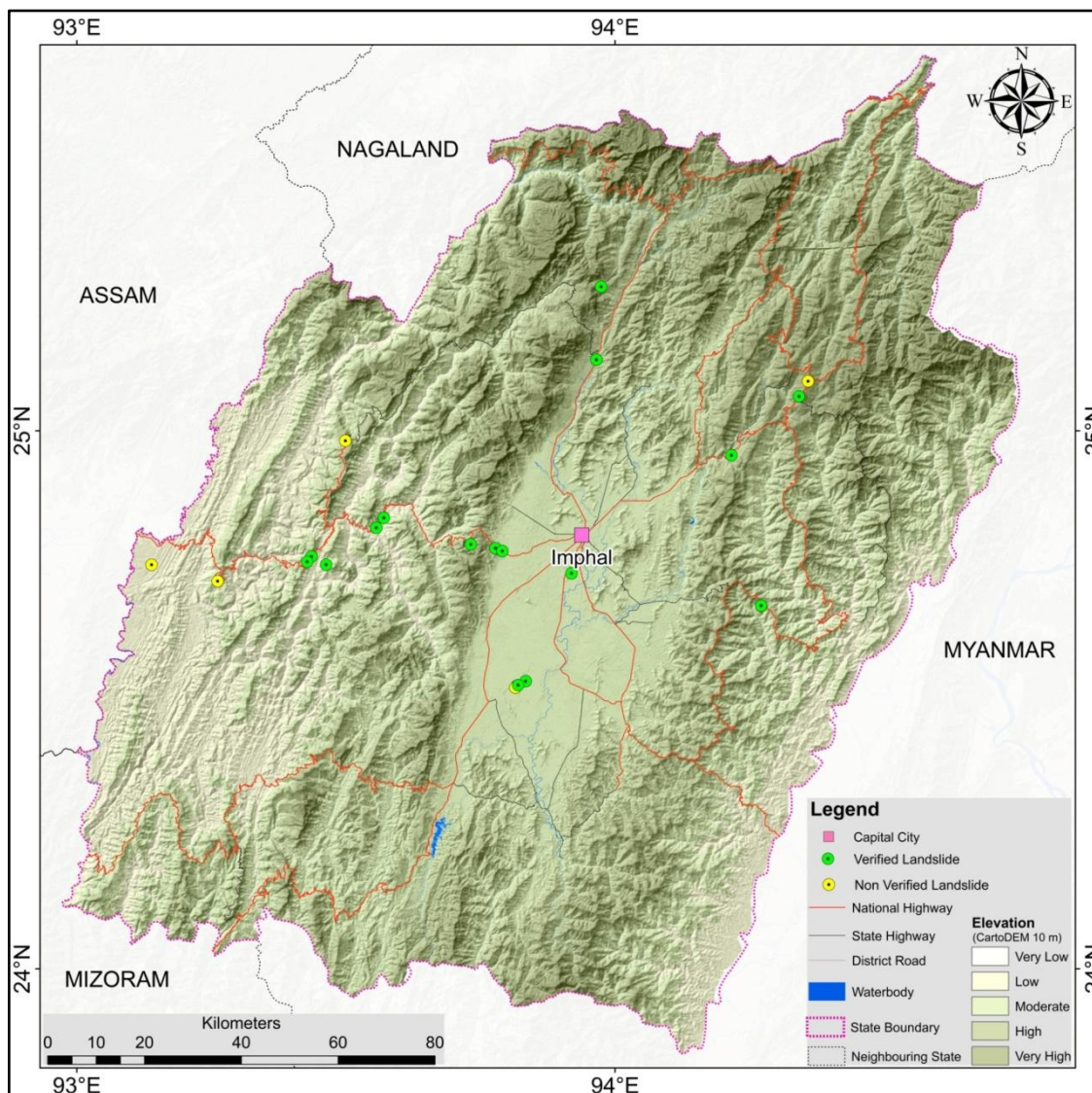
- A total of 15 landslide incidents recorded in Assam in the year 2024.
- The first landslide incident for the year was recorded on 1st May 2024 and the latest event was recorded on 10th August 2024.
- A total of 10 casualties occurred in Assam in 2024.
- Landslide in Assam caused lot of damages to properties and injuries to people.
- The spatial distributions and details of the landslides are shown in Map 3 and Annexure I(II) respectively.



Map 3 Landslide incidents in Assam, 2024

4.3 Highlights of Landslide incidents in Manipur

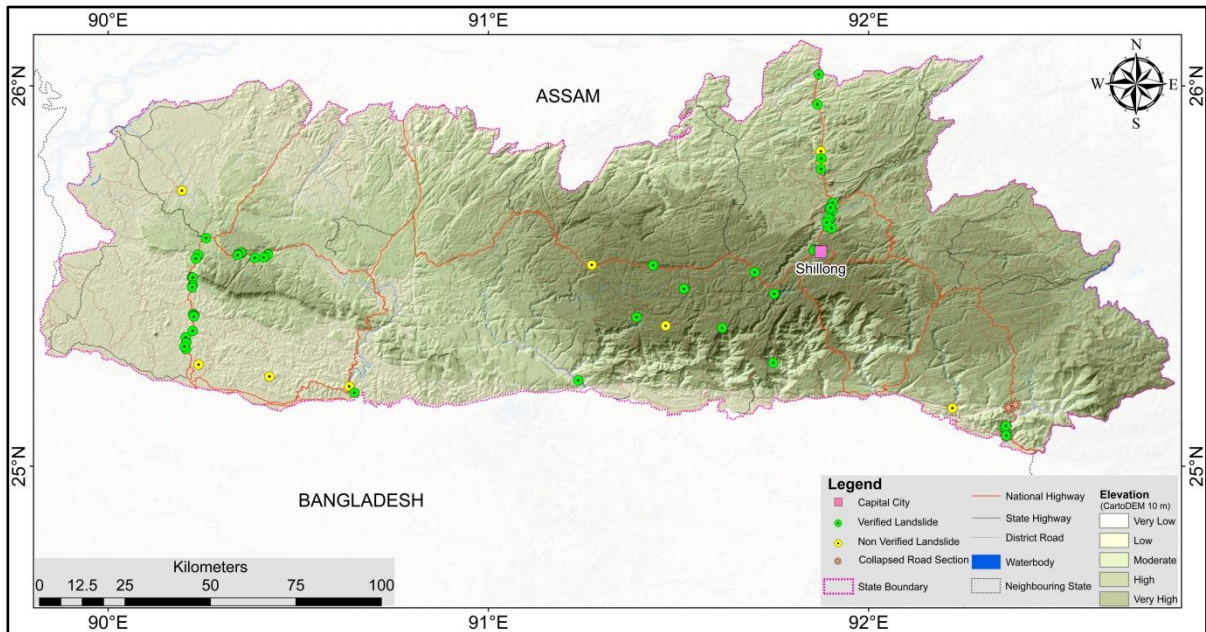
- A total of 22 landslide incidents recorded in Manipur in the year 2024.
- The first landslide incident for the year was recorded on 27th May 2024 and the latest event was recorded on 1st October 2024.
- A total of 5 casualties occurred in Manipur in 2024.
- Landslides caused lot of disruptions to traffic in Manipur at different occasions.
- The spatial distributions and details of the landslides are shown in Map 4 and Annexure I (III) respectively.



Map 4 Landslide incidents in Manipur, 2024

4.4 Highlights of Landslide incidents in Meghalaya

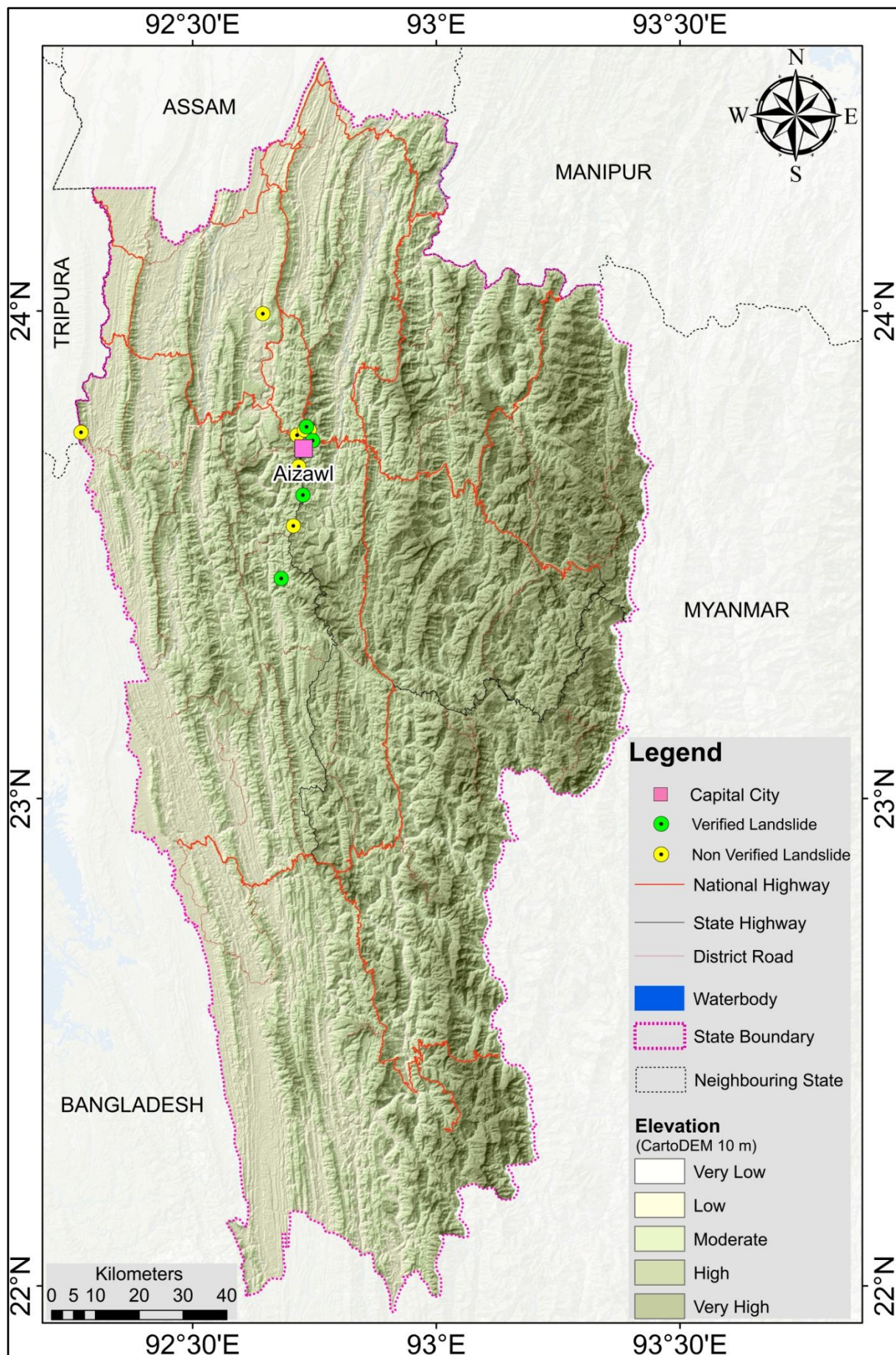
- A total of 94 landslide incidents recorded in Meghalaya in the year 2024.
- The active landslide near Sonapur Village/Tunnel remained a problem almost throughout the monsoon season while between February and August, at 4 instances it created major issues.
- The first landslide incident for the year was recorded on 3rd February 2024 and the latest event was recorded on 5th October 2024.
- 13 casualties were reported from Meghalaya in 2024.
- NH-6, NH-217 and SH-1 were some of the highways/road sections that were most affected in the state due to landslides.
- The spatial distributions and details of the landslides are shown in Map 5 and Annexure I(IV) respectively



Map 5 Landslide incidents in Meghalaya, 2024

4.5 Highlights of Landslide incidents in Mizoram

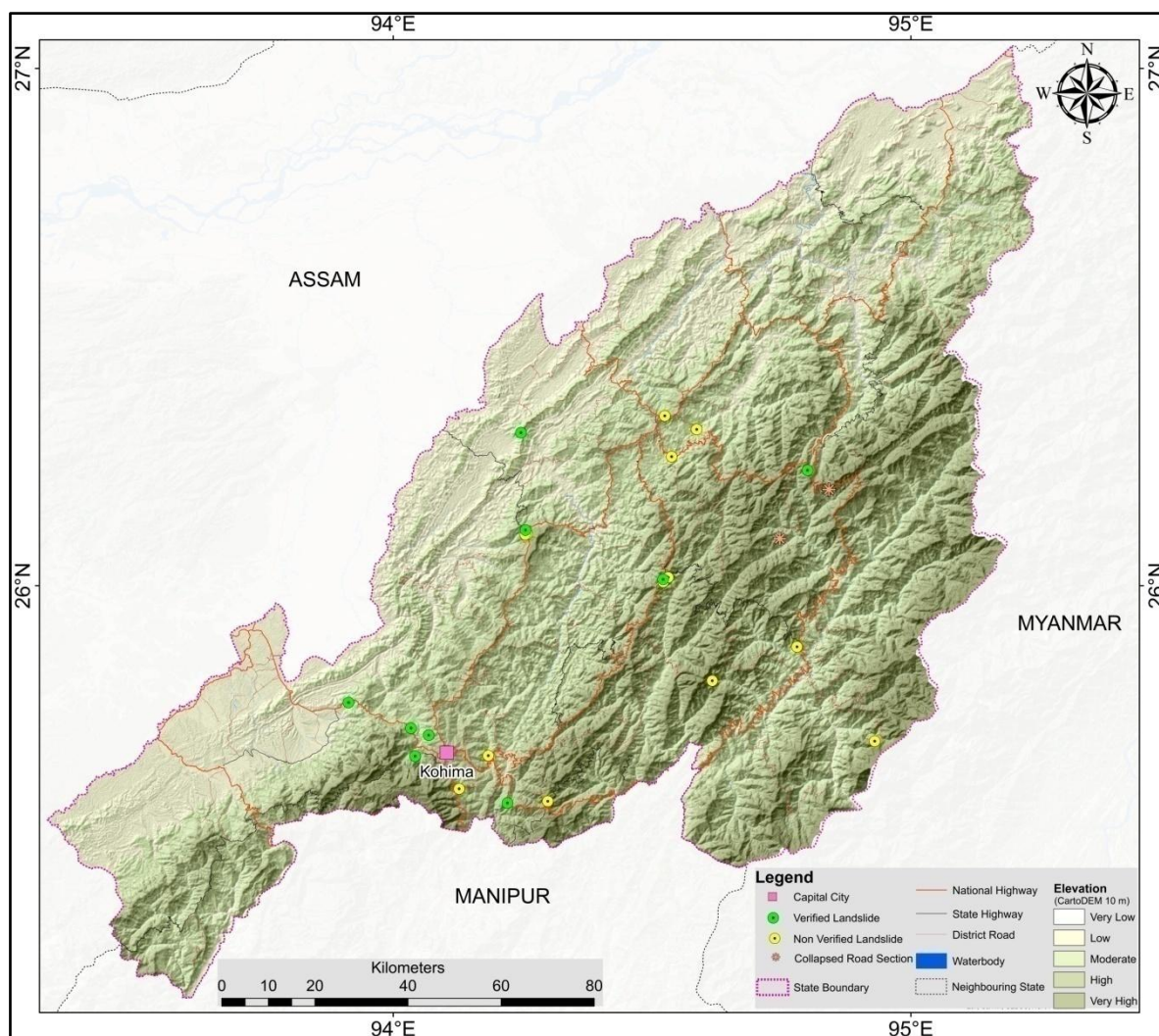
- 12 landslide incidents recorded in Mizoram. The first landslide incident was recorded on 28th May 2024 and the latest event was recorded on 28th August 2024. The 28th May landslide was the effect of Cyclone Remal which claimed several lives in the state.
- Hlimen landslide near stone mine in Southern Aizawl was a very tragic incident which claimed lives of 28 mine workers in Mizoram in 2024.
- NH-54 was the major road sections affected by landslides in Mizoram in 2024.
- The spatial distributions and details of the landslides are shown in Map 6 and Annexure I(V) respectively



Map 6 Landslide incidents in Mizoram, 2024

4.6 Highlights of Landslide incidents in Nagaland

- A total of 26 landslide incidents recorded in Nagaland in 2024.
- The first landslide incident for the year was recorded on 28th May 2024 and the last event was recorded on 4th October 2024.
- 7 casualties were reported from Nagaland in 2024.
- NH 2, NH 29 and NH 202 were the most affected roads due to landslide in 2024.
- The spatial distributions and details of the landslides are shown in Map 7 and Annexure I(VI) respectively.

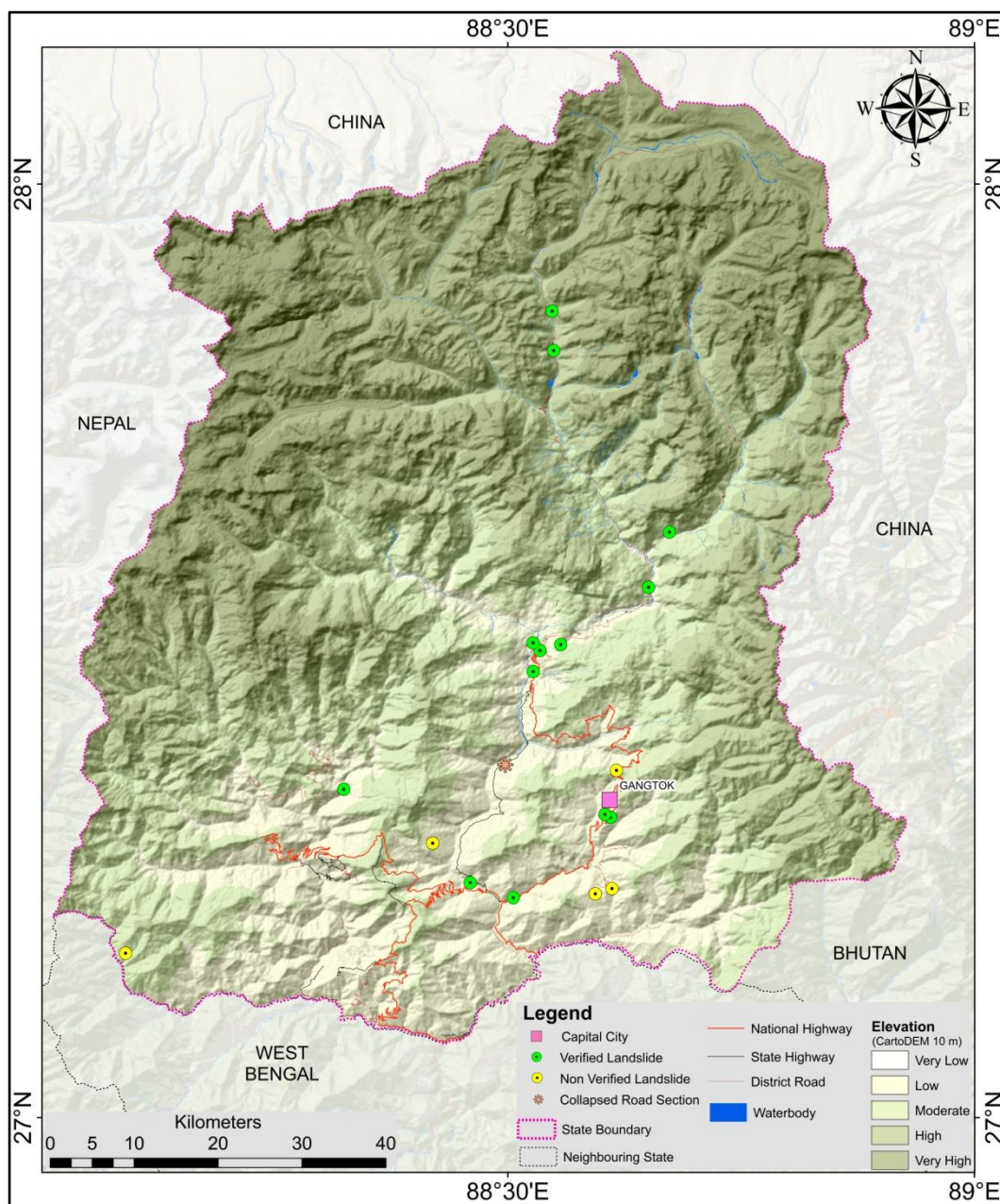


Map 7 Landslide incidents in Nagaland, 2024

4.7 Highlights of Landslide incidents in Sikkim

- 19 landslide incidents recorded in Sikkim in 2024. The first landslide incident for the year was recorded on 17th May 2024 and the latest event was recorded on 30th September 2024.
- 13 casualties were reported from Sikkim in 2024.

- Teesta Stage 5 Hydro Dam was damaged due to the landslide event that triggered on 20th August, 2024.
- NH-10 was among the major road sections affected by landslide.
- The spatial distributions and details of the landslides are shown in Map 8 and Annexure I(VII) respectively.

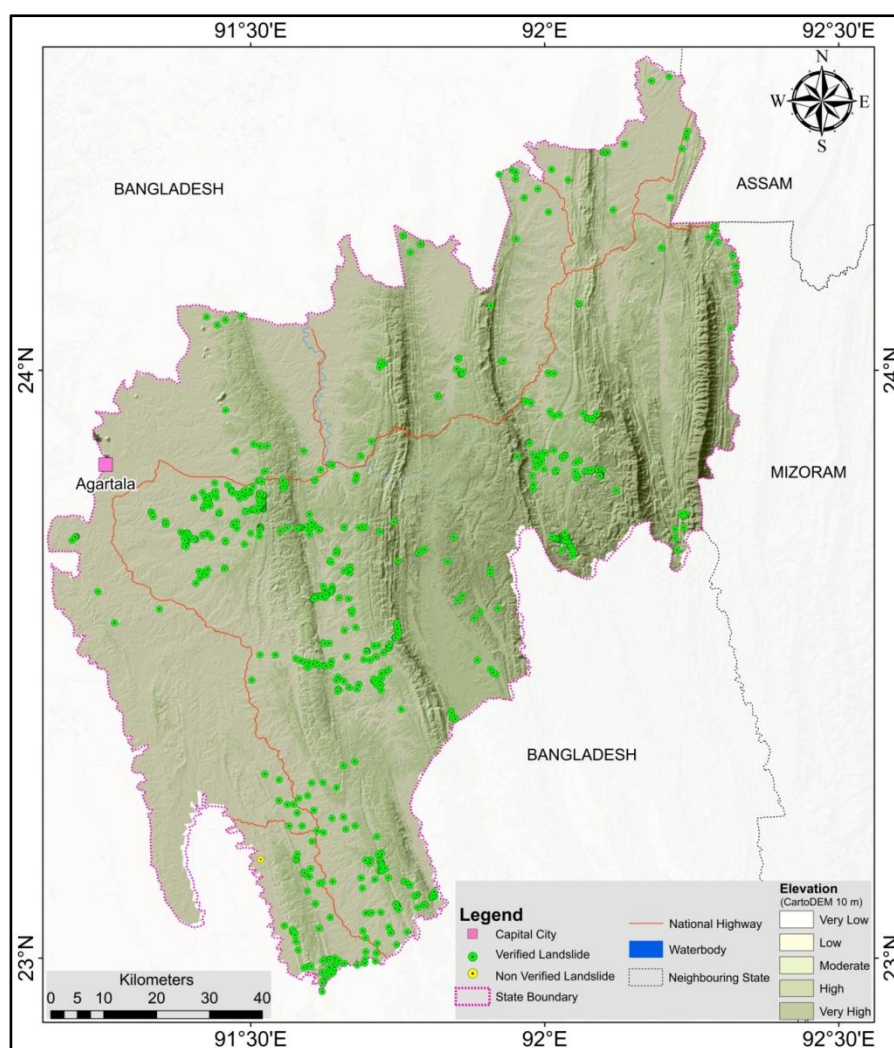


Map 8 Landslide incidents in Sikkim, 2024

4.8 Highlights of Landslide incidents in Tripura

- 920 landslide incidents recorded in Tripura in the year 2024.
- The first landslide incident for the year was recorded on 15th June 2024 and the latest event was recorded on 5th September 2024.

- Out of these 920 landslides, 914 landslides were reported by Tripura SDMA following the heavy rainfall incident during 19-23 August, 2024 (Tripura SDMA). Also, a separate report was prepared for this incident. Overall, more than 2000 incidents were reported by Tripura SDMA due to flood and landslide in the states.
- 11 casualties were reported from Tripura in 2024 due to landslide.
- In August 2024, the state of Tripura witnessed unprecedented heavy rainfall that induced severe flooding and landslides across the state. A record-breaking 288.8 mm rainfall was recorded in the state on August 20, severely affecting most of the infrastructures and all the river systems. All the districts received heavy rainfall over a period on one week. In South Tripura, 375.8 mm and 324.4 mm of rain were recorded in Bagafa and Belonia, respectively. Amarpur in Gomati district received 307.1 mm (Seeds India).
- The spatial distributions and details of the landslides are shown in Map 9 and Annexure I(VIII) respectively.



Map 9 Landslide incidents in Tripura, 2024

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6. ANNEXURES

ANNEXURE-I: LANDSLIDE INCIDENT REPORTS

Annexure-I(I): Arunachal Pradesh

S. No	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_m m)
1	Newspapers & Media/2024/Aarunachal Pradesh/2	Newspapers & Media	Dibang Valley	Hunli and Anini	Between Hunli and Anini	NH 313	28.233285	95.777581	24-04-2024	5:00 PM	3	0	31.155
2	Newspapers & Media/2024/Arunachal Pradesh/6	Newspapers & Media	Kurung Kumey	Pach Nallah Landslide	Between Leel village and Tagum camp/6 kilo under Sangram circle	Road	27.78598	93.542196	16-05-2024	Morning	3	0	1.8
3	Newspapers & Media/2024/Arunachal Pradesh/52	Newspapers & Media	Upper Siang	Bomdo Landslide	Between Ramsing and Bomdo		28.748529	94.910333	15-06-2024		3	0	23.9
4	Newspapers & Media/2024/Arunachal Pradesh/62	Newspapers & Media	Papum Pare	Karsingsa Landslide	Between Nirjuli and Banderdewa	NH 415	27.121869	93.783419	18-06-2024		3	0	47.33
5	Newspapers & Media/2024/Arunachal Pradesh/66	Newspapers & Media	Papum Pare	Itanagar Landslide	Itanagar City Area	NH 415	27.099654	93.62708	23-06-2024	10:30 AM	2	0	7.91
6	Newspapers & Media/2024/Arunachal Pradesh/67	Newspapers & Media	Changlang	Rangfra Landslide	Near Rangfra Mandir, Changlang		27.141177	95.746747	16-06-2024		3	0	9.29
7	Newspapers & Media/2024/Arunachal Pradesh/68	Newspapers & Media	West Kameng	Tippi Landslide	Road connecting Bomdila-Seppa-Tawang.		27.03649	92.605617	26-06-2024		3	0	163.89
8	Newspapers & Media/2024/Arunachal Pradesh/69	Newspapers & Media	Lower Siang	Siji Landslide	Likabali-Basar-Daporijo Road	NH 13	27.923784	94.711676	25-06-2024		3	0	276.6
9	Newspapers & Media/2024/Arunachal Pradesh/71	Newspapers & Media	Siang District	Tarak Landslide	Pasighat-Pangin-Aalo road	NH 13	28.226541	94.883203	29-06-2024		3	0	173.13
10	Newspapers & Media/2024/Arunachal Pradesh/72	Newspapers & Media	Papum Pare	Jullang Landslide	Ganga Jullang Road, Itanagar		27.07383	93.6327	29-06-2024		3	0	105.53
11	Newspapers & Media/2024/Arunachal Pradesh/73	Newspapers & Media	West Siang	Tatu Landslide	Aalo Mechuka Road		28.519848	94.361341	29-06-2024		2	0	543.15
12	Newspapers & Media/2024/Arunachal Pradesh/75	Newspapers & Media	Lohit	Demwe Landslide	Demwe-Brahmakund Tri Junction to Parashuram Kund-Wakro Road	NH 13	27.886843	96.374843	29-06-2024		3	0	339.6

S. No	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_m m)
13	Newspapers & Media/2024/Arunachal Pradesh/76	Newspapers & Media	Anjaw	Hayuliang Landslide	RamMandir-Tidding-Khupa-Hayuliang-Hawai Road	NH 113	28.06798	96.565515	29-06-2024		2	0	386.8
14	Newspapers & Media/2024/Arunachal Pradesh/78	Newspapers & Media	East Kameng	Bana Landslide	Bana Chepa Road	NH 13	27.30978	92.958741	30-06-2024		3	0	102.74
15	Newspapers & Media/2024/Arunachal Pradesh/79	Newspapers & Media	Upper Siang	Jengging Landslide	Pasighat - Mariyang-Yingkiong Road		28.676866	94.964479	30-06-2024		3	0	172.54
16	Newspapers & Media/2024/Arunachal Pradesh/80	Newspapers & Media	Siang District	Babuk Landslide	highway between Babuk and Kebang Village	NH 13	28.142213	95.092954	29-06-2024		3	0	197.4
17	Newspapers & Media/2024/Arunachal Pradesh/84	Newspapers & Media	Papum Pare	Division IV Landslide	Division IV Landslide area, Itanagar		27.090926	93.605705	01-07-2024		3	0	473.2
18	Newspapers & Media/2024/Arunachal Pradesh/85	Newspapers & Media	KurungKumey	Rengchi Landslide	Rengchi area in Kurung		27.895164	93.430041	30-06-2024		3	0	453.17
19	Newspapers & Media/2024/Arunachal Pradesh/105	Newspapers & Media	Papum Pare	Baat Landslide	Itanagar-Baat Road		27.089317	93.591888	01-07-2024		3	0	179.7
20	Newspapers & Media/2024/Arunachal Pradesh/106	Newspapers & Media	Siang District	Pangin Landslide	Lokpeng and Pangin Tri-junction		28.233305	94.98636	04-07-2024	Morning	2	0	2.999
21	Newspapers & Media/2024/Arunachal Pradesh/117	Newspapers & Media	Siang District	Babuk Landslide	highway between Babuk and Kebang Village	NH 13	28.142494	95.093074	06-07-2024	1:00 AM	3	0	24.3
22	Newspapers & Media/2024/Arunachal Pradesh/118	Newspapers & Media	Papum Pare	Karsingsa Landslide	Between Nirjuli and Banderdewa	NH 415	27.121839	93.783633	05-07-2024	Morning	3	0	106.57
23	Newspapers & Media/2024/Arunachal Pradesh/119	Newspapers & Media	Shi Yomi	Shi Yomi Landslide			28.697524	94.316142	05-07-2024		2	1	123.13
24	Newspapers & Media/2024/Arunachal Pradesh/120	Newspapers & Media	Anjaw	Mompani Landslide	Tezu-Hayuliang road		28.047143	96.439961	05-07-2024		3	0	108.67
25	Newspapers & Media/2024/Arunachal Pradesh/121	Newspapers & Media	KraDaadi	Palin Landslide	Palin-Taraklengdi Road		27.72775	93.636163	05-07-2024		2	0	5.18
26	Newspapers & Media/2024/Arunachal Pradesh/127	Newspapers & Media	West Kameng	Palizi Landslide	Between Rupa and Seppa		27.307091	92.792901	14-07-2024		3	0	13.1
27	Newspapers & Media/2024/Arunachal Pradesh/134	Newspapers & Media	Papum Pare	Holongi Landslide	Between Holongi and Itanagar	NH 415	27.031002	93.621829	22-07-2024		3	0	78.83
28	Newspapers & Media/2024/Arunachal Pradesh/176	Newspapers & Media	Tawang	Bramadungchung Landslide	Near Bramadungchung Ani Gonpa		27.611051	91.857654	07-09-2024	4:00 AM	2	0	9.57

S. No.	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_mm)
29	Newspapers & Media/2024/Arunachal Pradesh/186	Newspapers & Media	West Kameng	Kaspi Landslide	Kaspi Vilage, Near Nag Mandir		27.203922	92.562963	04-10-2024		3	0	137.85

Annexure-I (II): Assam

S. No.	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_mm)
1	Newspapers & Media/2024/Assam/4	Newspapers & Media	Dima Hasao	Jatinga Landslide	Between Jatinga and Harangajao Station		25.109872	92.891914	01-05-2024		3	0	15.646
2	Newspapers & Media/2024/Assam/5	Newspapers & Media	Cachar	Damcherra Landslide	Badarpur Pahar line		25.019471	92.76249	03-05-2024		3	0	9.759
3	Newspapers & Media/2024/Assam/11	Newspapers & Media	Tinsukia	Tikok Landslide	Situated between Bargolai and Namdang under the Tikok colliery in Ledo		27.282038	95.738226	26-05-2024	12:30 AM	2	0	21.025
4	Newspapers & Media/2024/Assam/12	Newspapers & Media	Dhemaji	Subansiri Landslide	NHPC Lower Subansiri Hydroelectricity Project in Gogamukh		27.532524	94.263033	27-05-2024	Day Time	2	1	50.167
5	Newspapers & Media/2024/Assam/32	Newspapers & Media	Kamrup Metro	Krishna Nagar Landslide	Lal Ganesh Area		26.144954	91.747639	29-05-2024		2	1	206.555
6	Newspapers & Media/2024/Assam/36	Newspapers & Media	Kamrup Metro	Katahbari Landslide	Masjid Road No. 1, Katahbari		26.125215	91.71827	03-06-2024	Daytime	3	1	70.83
7	Newspapers & Media/2024/Assam/37	Newspapers & Media	Kamrup Metro	Gramya Nagar Landslide	Gramya Nagar Path, Pandu		26.160366	91.698208	03-06-2024	Daytime	2	0	72.56
8	Newspapers & Media/2024/Assam/38	Newspapers & Media	Dima Hasao	Mahur Landslide	Mahur Lasang Road	NH 27	25.179895	93.116465	30-05-2024		3	0	154.662
9	Newspapers & Media/2024/Assam/39	Newspapers & Media	Dima Hasao	Lungkhok Landslide	Road side, Lungkhok		25.202045	93.01028	30-05-2024		3	0	511.005
10	Newspapers & Media/2024/Assam/40	Newspapers & Media	Dima Hasao	Bethel Landslide	Road side, Bethel		25.1934	93.020431	30-05-2024		3	0	373.117
11	Newspapers & Media/2024/Assam/41	Newspapers & Media	Kamrup Metro	Rudra Nagar Landslide	Chandmari, Guwahati		26.190964	91.775725	30-05-2024	1:30 AM	2	0	44.782
12	Newspapers & Media/2024/Assam/63	Newspapers & Media	Karimganj	Bendargool Landslide	Between Gainachora-Bendargool village		24.787043	92.523786	19-06-2024	12:45 AM	2	5	199.48

S. No.	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_mm)
13	Newspapers & Media/2024/Assam/115	Newspapers & Media	Kamrup Metro	8 th Mile Landslide	8 th Mile Area, Jorabat		26.111529	91.846223	04-07-2024	Night	2	2	35.38
14	Newspapers & Media/2024/Assam/125	Newspapers & Media	Kamrup Metro	Bonda Landslide	Guwahati's Bonda locality, Amgaon		26.179508	91.829241	12-07-2024		2	0	182.02
15	Newspapers & Media/2024/Assam/154	Newspapers & Media	Dima Hasao	Damchera Landslide	SilcharHaflong Road section	NH 27	25.006926	92.747317	10-08-2024		3	0	0

Annexure-I (III): Manipur

S. No.	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_mm)
1	Newspapers & Media/2024/Manipur/16	Newspapers & Media	Ukhrul	Kasom Landslide	Near Kasom Village		24.67507	94.27133	27-05-2024	10:30 AM	3	0	89.176
2	Newspapers & Media/2024/Manipur/30	Newspapers & Media	Imphal	Imphal Landslide	Imphal Jiribam Road	NH 37	24.78167	93.7777	28-05-2024		3	0	347.347
3	Newspapers & Media/2024/Manipur/31	Newspapers & Media	Senapati	Senapati Landslide	Senapati Thonglang Road		25.26763	93.97409	28-05-2024		2	1	271.775
4	Newspapers & Media/2024/Manipur/98	Newspapers & Media	Ukhrul	MongkotChep u Landslide	Imphal Ukhrul Road	NH 202	24.95445	94.21625	02-07-2024	9:00 AM	2	0	428.38
5	Newspapers & Media/2024/Manipur/100	Newspapers & Media	Ukhrul	Kazipphung Landslide	Imphal-Ukhrul Road	NH 202	25.06391	94.34178	03-07-2024	6:00 AM	2	0	37.95
6	Newspapers & Media/2024/Manipur/114	Newspapers & Media	Kangpokpi	Kangpokpi Landslide	Between Daili village and Yaikongpao village	NH 2	25.13191	93.96529	04-07-2024	11:00 PM	2	0	32.18
7	Newspapers & Media/2024/Manipur/139	Newspapers & Media	Tamenglong	Dimthanlong Landslide	Dimthanlong village, Tamenglong Ward No. 3		24.98131	93.49901	29-07-2024		2	2	1.19
8	Newspapers & Media/2024/Manipur/140	Newspapers & Media	Imphal East	Sinam Landslide	Along NH-37 near K. Sinam village	NH 37	24.7764	93.79057	29-07-2024		2	0	11.86
9	Newspapers & Media/2024/Manipur/141	Newspapers & Media	Tamenglong	Rengpang Landslide	Between Nungba and Rengpang villages	NH 37	24.76632	93.43601	29-07-2024		2	0	0
10	Newspapers & Media/2024/Manipur/142	Newspapers & Media	Tamenglong	Taodaijang Landslide	Between Taodaijang and Rengpang village		24.75121	93.46316	29-07-2024		2	0	0
11	Newspapers & Media/2024/Manipur/143	Newspapers & Media	Bishnupur	Thanga Part I Landslide	Thanga Part I Gram Panchayat		24.53455	93.83401	01-08-2024		3	0	29.43

S. No.	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_mm)
12	Newspapers & Media/2024/Manipur/144	Newspapers & Media	Bishnupur	Thanga Part II Landslide	Thanga Part II Gram Panchayat		24.52563	93.81729	01-08-2024		2	0	30.67
13	Newspapers & Media/2024/Manipur/145	Newspapers & Media	Bishnupur	Thanga Ngaram Landslide	Thanga NgaramChinkha Region		24.5226	93.81477	01-08-2024		2	0	29.43
14	Newspapers & Media/2024/Manipur/146	Newspapers & Media	Bishnupur	Oinam Landslide	OinamLeikai		24.73502	93.919	01-08-2024		2	0	12.63
15	Newspapers & Media/2024/Manipur/147	Newspapers & Media	Bishnupur	Heisnam Landslide	Thanga Region		24.52711	93.82001	01-08-2024		2	0	29.43
16	Newspapers & Media/2024/Manipur/150	Newspapers & Media	Jiribam	JiribamSonapur Landslide	Sonapur Gram Panchayat, Jiribam		24.75096	93.13879	06-08-2024	3:30 AM	2	1	71.32
17	Newspapers & Media/2024/Manipur/151	Newspapers & Media	Churachandpur	Nungba Landslide	Near Nungba region	NH 37	24.75648	93.42788	07-08-2024	Early Hours	3	0	34.76
18	Newspapers & Media/2024/Manipur/152	Newspapers & Media	Tamenglong	Kaimai Landslide	Kaimai Village area	NH 37	24.72047	93.26164	07-08-2024		3	0	40.56
19	Newspapers & Media/2024/Manipur/157	Newspapers & Media	Ukhrul	Hamleikhong Landslide	Hamleikhong Area		25.092	94.35854	12-08-2024	5:00 PM	2	1	26.8
20	Newspapers & Media/2024/Manipur/160	Newspapers & Media	Kangpokpi	K. Sinam Landslide	K. Sinam Village in Kangpokpi District along NH-37	NH 37	24.78893	93.73187	18-08-2024		2	0	104.36
21	Newspapers & Media/2024/Manipur/161	Newspapers & Media	Noney	Awangkhum Landslide	Noney District along NH-37	NH 37	24.81961	93.55647	18-08-2024		3	0	104.36
22	Newspapers & Media/2024/Manipur/182	Newspapers & Media	Noney	AwangkhumLangkhung Landslide	Between Awangkhum and Langkhong	NH 37	24.83803	93.5706	01-10-2024	Morning	2	0	2.21

Annexure-I (IV): Meghalaya

S. No.	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_mm)
1	Newspapers & Media/2024/Meghalaya/1	Newspapers & Media	East Jaintia Hills	Sonapur Tunnel Landslide	Near Sonapur Tunnel	NH 6	25.105844	92.36018	03-02-2024	4:00 AM	3	0	2.081
2	Newspapers & Media/2024/Meghalaya/3	Newspapers & Media	East Jaintia Hills	Sonapur Landslide	Near Sonapur Tunnel	NH 6	25.102667	92.358006	30-04-2024	6:00 PM	3	0	3.825

S. No.	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_mm)
3	Newspapers & Media/2024/Meghalaya/7	Newspapers & Media	East Khasi Hills	3rd Mile Landslide	Near St Gabriell Primary School		25.569287	91.856166	16-05-2024		3	0	7.69
4	Newspapers & Media/2024/Meghalaya/8	Newspapers & Media	East Khasi Hills	Borma Landslide	Near Borma College, 3rd Mile		25.568237	91.856614	16-05-2024		3	0	7.69
5	Newspapers & Media/2024/Meghalaya/10	Newspapers & Media	East Khasi Hills	Shillong Landslide	Near New Assembly building, Meghalaya		25.568556	91.873806	17-05-2024		2	0	97.553
6	Newspapers & Media/2024/Meghalaya/17	Newspapers & Media	East Jaintia Hills	Sonapur Tunnel Landslide	Near Sonapur Tunnel	NH 6	25.106205	92.360977	27-05-2024	10:00 AM	3	0	227.273
7	Newspapers & Media/2024/Meghalaya/18	Newspapers & Media	East khasi Hills	Weilong Landslide	East Khasi Hills		25.363389	91.614595	28-05-2024	Morning	2	1	278.215
8	Newspapers & Media/2024/Meghalaya/19	Newspapers & Media	East khasi Hills	Mawlai Landslide	East Khasi Hills		25.62836	91.893134	28-05-2024	4:00 AM	2	0	292.477
9	Newspapers & Media/2024/Meghalaya/20	Newspapers & Media	East khasi Hills	Mawkynroh Landslide	East Khasi Hills		25.625808	91.902597	28-05-2024	5:00 AM	2	0	293.352
10	Newspapers & Media/2024/Meghalaya/21	Newspapers & Media	East khasi Hills	Mawphland Landslide	Mawphland Block		25.453314	91.751767	28-05-2024	Morning	3	0	302.408
11	Newspapers & Media/2024/Meghalaya/22	Newspapers & Media	East khasi Hills	Rilbong Landslide	Near Rilbong Beat House		25.567062	91.871489	28-05-2024	Morning	2	0	115.358
12	Newspapers & Media/2024/Meghalaya/23	Newspapers & Media	East khasi Hills	Lumparing Landslide	Lower Lumparing		25.560962	91.881821	28-05-2024	Morning	2	0	115.358
13	Newspapers & Media/2024/Meghalaya/24	Newspapers & Media	South West Khasi Hills	13 KM Landslide	MR Road		25.39192932	91.3896869	28-05-2024	Morning	2	0	73.799
14	Newspapers & Media/2024/Meghalaya/35	Newspapers & Media	East Khasi Hills	Nongpriang Landslide	Nongpriang village under Shella Block		25.271534	91.748951	30-05-2024		3	4	564.639
15	Newspapers & Media/2024/Meghalaya/42	Newspapers & Media	East Jaintia Hills	Lumshnong Landslide	Along Lumshnong Road Side	NH 6	25.15457	92.368435	30-05-2024		3	0	599.519
16	Newspapers & Media/2024/Meghalaya/44	Newspapers & Media	East Jaintia Hills	Lumshnong 2 Landslide	Lumshnong area along NH 6	NH 6	25.162582	92.38607	15-06-2024	12:25 PM	3	0	246
17	Newspapers & Media/2024/Meghalaya/53	Newspapers & Media	South West Khasi Hills	Lummawbah Landslide	South West Khasi Hills		25.568354	91.866635	15-06-2024		2	0	140

S. No.	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_mm)
18	Newspapers & Media/2024/Meghalaya/55	Newspapers & Media	West Khasi Hills	Markasa Landslide	Markasa village under Mawthadraishan block		25.52819	91.433141	17-06-2024		3	0	312
19	Newspapers & Media/2024/Meghalaya/57	Field Survey	Ri Bhoi	Ri Palei Landslide	Ri Bhoi region	NH 6	25.82781	91.874287	14-06-2024		2	0	107
20	Newspapers & Media/2024/Meghalaya/58	Field Survey	Ri Bhoi	Umsamlem Landslide	Ri Bhoi region	NH 6	25.800588	91.875187	14-06-2024		2	0	107
21	Newspapers & Media/2024/Meghalaya/59	Field Survey	Ri Bhoi	Umiam Landslide	Ri Bhoi region	NH 6	25.679203	91.899853	14-06-2024		2	0	234
22	Newspapers & Media/2024/Meghalaya/60	Newspapers & Media	South West Khasi Hills	Mawkyrwat Landslide	Mawkyrwat block		25.368971	91.466408	15-06-2024		2	0	272.9
23	Newspapers & Media/2024/Meghalaya/61	Newspapers & Media	South West Khasi Hills	Ranikor Landslide	Ranikor Block		25.225818	91.236212	15-06-2024		2	0	147.44
24	Newspapers & Media/2024/Meghalaya/64	Field Survey	East Khasi Hills	Pomlum Landslide	Pomlum village in Laitlyngkot		25.518951	91.828582	19-06-2024		2	0	349
25	Newspapers & Media/2024/Meghalaya/65	Newspapers & Media	East Jaintia Hills	Karkhana Landslide	Karkhana Village		25.152774	92.220332	17-06-2024		2	0	276
26	Newspapers & Media/2024/Meghalaya/81	Field Survey	East Khasi Hills	Sohiong Landslide	Mawmaram Road		25.510164	91.700373	28-06-2024		1	0	43.61
27	Newspapers & Media/2024/Meghalaya/82	Field Survey	Ri Bhoi	Umiam 2 Landslide	Near Umiam Fire Station	NH 6	25.669593	91.903928	30-06-2024		3	0	114.34
28	Newspapers & Media/2024/Meghalaya/83	Field Survey	Ri Bhoi	Umiam Dam Landslide	Near Umiam Dam	NH 6	25.653138	91.898077	01-07-2024		3	0	494.05
29	Newspapers & Media/2024/Meghalaya/87	Field Survey	Ri Bhoi	Umbang 1 Landslide	GS Road, Ri Bhoi	NH 6	25.682492	91.900697	30-06-2024		3	0	129.28
30	Newspapers & Media/2024/Meghalaya/88	Field Survey	Ri Bhoi	NEPA Landslide	GS Road, Ri Bhoi	NH 6	25.676048	91.897497	30-06-2024		3	0	129.28
31	Newspapers & Media/2024/Meghalaya/89	Field Survey	Ri Bhoi	Umiam 3 Landslide	GS Road, Ri Bhoi	NH 6	25.675099	91.898378	30-06-2024		3	0	114.34
32	Newspapers & Media/2024/Meghalaya/90	Field Survey	Ri Bhoi	Umbang 2 Landslide	GS Road, Ri Bhoi	NH 6	25.686237	91.902376	30-06-2024		3	0	114.34
33	Newspapers & Media/2024/Meghalaya/91	Field Survey	Ri Bhoi	Umbang 3 Landslide	GS Road, Ri Bhoi	NH 6	25.686801	91.903145	30-06-2024		3	0	114.34

S. No.	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_mm)
34	Newspapers & Media/2024/Meghalaya/92	Field Survey	Ri Bhoi	Sumer Landslide	GS Road, Ri Bhoi	NH 6	25.692127	91.906341	30-06-2024		3	0	129.28
35	Newspapers & Media/2024/Meghalaya/93	Field Survey	Ri Bhoi	Umbang 4 Landslide	GS Road, Ri Bhoi	NH 6	25.67939	91.899541	30-06-2024		3	0	8.6
36	Newspapers & Media/2024/Meghalaya/96	Newspapers & Media	West Khasi Hills	Tiehsaw Landslide	Nongstoin-Shillong Highway		25.529008	91.271797	02-07-2024		2	0	170.76
37	Newspapers & Media/2024/Meghalaya/97	Newspapers & Media	West Khasi Hills	Pariong Landslide	Lawbyrtun-Rngikseh road under Mawthadraishan Block		25.466183	91.514339	02-07-2024		3	0	162.64
38	Newspapers & Media/2024/Meghalaya/107	Newspapers & Media	South Garo Hills	Baghmara Landslide	Near Baghmara SBI		25.193381	90.647418	04-07-2024	Morning	2	0	4.57
39	Newspapers & Media/2024/Meghalaya/108	Newspapers & Media	South Garo Hills	Baghmara 2 Landslide	Near Baghmara Circuit House		25.210055	90.633676	04-07-2024		2	0	13.29
40	Newspapers & Media/2024/Meghalaya/109	Field Survey	Ri Bhoi	Umiam 4 Landslide	GS Road, Ri Bhoi	NH 6	25.649229	91.898983	02-07-2024		3	0	263.53
41	Newspapers & Media/2024/Meghalaya/110	Field Survey	Ri Bhoi	Umiam 5 Landslide	GS Road, Ri Bhoi	NH 6	25.650726	91.898692	02-07-2024		3	0	263.53
42	Newspapers & Media/2024/Meghalaya/111	Field Survey	Ri Bhoi	Umiam View Point Landslide	GS Road, Ri Bhoi	NH 6	25.651025	91.893994	02-07-2024		3	0	263.53
43	Newspapers & Media/2024/Meghalaya/112	Field Survey	East khasi Hills	Mawlai 2 Landslide	GS Road, East Khasi Hills	NH 6	25.641388	91.889399	02-07-2024		3	0	263.53
44	Newspapers & Media/2024/Meghalaya/113	Field Survey	East khasi Hills	Mawlai 3 Landslide	GS Road, East Khasi Hills	NH 6	25.640973	91.889427	02-07-2024		3	0	50.63
45	Newspapers & Media/2024/Meghalaya/122	Newspapers & Media	East Jaintia Hills	Kuliang Landslide	Kuliang section of National Highway 06	NH 6	25.08048	92.362414	12-07-2024	Morning	3	0	102.95
46	Newspapers & Media/2024/Meghalaya/126	Newspapers & Media	West Garo Hills	Dadenggre Landslide	Dadenggre subdivision, Tura		25.724652	90.193856	13-07-2024		2	0	4.07
47	Newspapers & Media/2024/Meghalaya/129	Field Survey	Ri Bhoi	Pahamrinai Landslide	GS Road, PahamrinaiVillahge, Ri Bhoi		25.809077	91.875303	14-07-2024		3	0	0.899
48	Newspapers & Media/2024/Meghalaya/130	Field Survey	Ri Bhoi	Umran Landslide	GS Road, Umran Villahge, Ri Bhoi		25.781194	91.874717	06-07-2024		3	0	28.96
49	Newspapers & Media/2024/Meghalaya/131	Newspapers & Media	Ri Bhoi	Maupur Landslide	Near Merrily Service Station, Ri Bhoi		26.029622	91.869058	18-07-2024		3	0	26.66

S. No.	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_mm)
50	Newspapers & Media/2024/Meghalaya/148	Newspapers & Media	East Jaintia Hills	Kuliang Landslide	Kuliang section of National Highway 06	NH 6	25.080525	92.362358	06-08-2024	Early Hours	3	0	9.4
51	Newspapers & Media/2024/Meghalaya/153	Newspapers & Media	East Jaintia Hills	Sonapur Tunnel Landslide	Near Sonapur Tunnel	NH 6	25.106369	92.361029	06-08-2024		3	0	5.14
52	Newspapers & Media/2024/Meghalaya/155	Newspapers & Media	East Jaintia Hills	Kuliang Landslide	Kuliang section of National Highway 06	NH 6	25.080525	92.362247	12-08-2024	Morning Hours	3	0	21.19
53	Newspapers & Media/2024/Meghalaya/159	Newspapers & Media	East Jaintia Hills	Kuliang Landslide	Kuliang section of National Highway 06	NH 6	25.080525	92.362123	17-08-2024	Afternoon	2	0	35.89
54	Newspapers & Media/2024/Meghalaya/175	Newspapers & Media	Ri Bhoi	Shangbangla Landslide	Shangbangla Village	NH 6	25.951567	91.865183	06-09-2024	8:00 PM	3	0	48.08
55	Newspapers & Media/2024/Meghalaya/178	Newspapers & Media	East Jaintia Hills	Kuliang Landslide	Kuliang section of National Highway 06	NH 6	25.080516	92.362183	30-09-2024		3	0	90.99
56	Newspapers & Media/2024/Meghalaya/184	Newspapers & Media	West Garo Hills	Koinadubi Landslide	Koinadubi Village, West Garo Hills		25.267236	90.237665	04-10-2024		2	1	189.37
57	Newspapers & Media/2024/Meghalaya/185	Newspapers & Media	South Garo Hills	Attesia Landslide	AttesiaSonmong, Gasuapara Region		25.236002	90.424197	05-10-2024		2	7	215.46
58	Newspapers & Media/2024/Meghalaya/187	Field Survey	West Garo Hills	Rongram Landslide	Near Rongram Nepali Baptist Church	NH 217	25.599743	90.257292	15-06-2024		1	0	125.69
59	Newspapers & Media/2024/Meghalaya/188	Field Survey	East Garo Hills	Mangrure Landslide 1	Tura Williamnagar Road	SH 1	25.556716	90.420247	15-06-2024		1	0	125.69
60	Newspapers & Media/2024/Meghalaya/189	Field Survey	East Garo Hills	Mangrure Landslide 2	Tura Williamnagar Road	SH 1	25.556503	90.420122	15-06-2024		1	0	125.69
61	Newspapers & Media/2024/Meghalaya/190	Field Survey	East Garo Hills	Mangrure Landslide 3	Tura Williamnagar Road	SH 1	25.556389	90.420051	15-06-2024		1	0	125.69
62	Newspapers & Media/2024/Meghalaya/191	Field Survey	East Garo Hills	Simsang Landslide 1	Tura Williamnagar Road	SH 1	25.549989	90.412596	15-06-2024		1	0	125.69
63	Newspapers & Media/2024/Meghalaya/192	Field Survey	East Garo Hills	Simsang Landslide 2	Tura Williamnagar Road	SH 1	25.55002	90.412686	15-06-2024		1	0	125.69

S. No.	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_mm)
64	Newspapers & Media/2024/Meghalaya/193	Field Survey	East Garo Hills	Simsang Landslide 3	Tura Williamnagar Road	SH 1	25.549016	90.408643	15-06-2024		1	0	125.69
65	Newspapers & Media/2024/Meghalaya/194	Field Survey	East Garo Hills	Durakalakgre Landslide	Tura Williamnagar Road	SH 1	25.547884	90.385679	15-06-2024		1	0	125.69
66	Newspapers & Media/2024/Meghalaya/195	Field Survey	West Garo Hills	Matchurigre Landslide 1	Tura Williamnagar Road	SH 1	25.561542	90.351459	15-06-2024		1	0	125.69
67	Newspapers & Media/2024/Meghalaya/196	Field Survey	West Garo Hills	Matchurigre Landslide 2	Tura Williamnagar Road	SH 1	25.561887	90.351379	15-06-2024		1	0	125.69
68	Newspapers & Media/2024/Meghalaya/197	Field Survey	West Garo Hills	Matchurigre Landslide 3	Tura Williamnagar Road	SH 1	25.561884	90.351379	15-06-2024		1	0	125.69
69	Newspapers & Media/2024/Meghalaya/198	Field Survey	West Garo Hills	Matchurigre Landslide 4	Tura Williamnagar Road	SH 1	25.561705	90.349382	15-06-2024		1	0	125.69
70	Newspapers & Media/2024/Meghalaya/199	Field Survey	West Garo Hills	Matchurigre Landslide 5	Tura Williamnagar Road	SH 1	25.557146	90.341414	15-06-2024		1	0	125.69
71	Newspapers & Media/2024/Meghalaya/200	Field Survey	West Garo Hills	Matchurigre Landslide 6	Tura Williamnagar Road	SH 1	25.55621	90.341198	15-06-2024		1	0	125.69
72	Newspapers & Media/2024/Meghalaya/201	Field Survey	West Garo Hills	Matchurigre Landslide 7	Tura Williamnagar Road	SH 1	25.555088	90.340616	15-06-2024		1	0	125.69
73	Newspapers & Media/2024/Meghalaya/202	Field Survey	West Garo Hills	Chasingre Landslide	Tura RongramDedenggre Road	NH 217	25.556378	90.236012	15-06-2024		1	0	125.69
74	Newspapers & Media/2024/Meghalaya/203	Field Survey	West Garo Hills	Rongkon Landslide	Near Rongkon Poultry Farm		25.545538	90.230769	15-06-2024		1	0	125.69
75	Newspapers & Media/2024/Meghalaya/204	Field Survey	West Garo Hills	Rengsipara Landslide 1	Paikan Tura Dalu Road	NH 217	25.308286	90.204561	15-06-2024		1	0	125.69
76	Newspapers & Media/2024/Meghalaya/205	Field Survey	West Garo Hills	Rengsipara Landslide 2	Paikan Tura Dalu Road	NH 217	25.308134	90.204644	15-06-2024		1	0	125.69
77	Newspapers & Media/2024/Meghalaya/206	Field Survey	West Garo Hills	Bolpuchring Landslide 1	Paikan Tura Dalu Road	NH 217	25.496547	90.220672	15-06-2024		1	0	125.69

S. No.	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_mm)
78	Newspapers & Media/2024/Meghalaya/207	Field Survey	West Garo Hills	Bolpuchring Landslide 2	Paikan Tura Dalu Road	NH 217	25.496702	90.221742	15-06-2024		1	0	125.69
79	Newspapers & Media/2024/Meghalaya/208	Field Survey	West Garo Hills	Bolpuchring Landslide 3	Paikan Tura Dalu Road	NH 217	25.496123	90.222287	15-06-2024		1	0	125.69
80	Newspapers & Media/2024/Meghalaya/209	Field Survey	West Garo Hills	Derengre Landslide	Paikan Tura Dalu Road	NH 217	25.47651	90.223143	15-06-2024		1	0	125.69
81	Newspapers & Media/2024/Meghalaya/210	Field Survey	West Garo Hills	Dinasagre Landslide 1	Paikan Tura Dalu Road	NH 217	25.470969	90.220833	15-06-2024		1	0	125.69
82	Newspapers & Media/2024/Meghalaya/211	Field Survey	West Garo Hills	Dinasagre Landslide 2	Paikan Tura Dalu Road	NH 217	25.469956	90.220708	15-06-2024		1	0	125.69
83	Newspapers & Media/2024/Meghalaya/212	Field Survey	West Garo Hills	Rongbretgre Landslide 1	Paikan Tura Dalu Road	NH 217	25.399736	90.223057	15-06-2024		1	0	125.69
84	Newspapers & Media/2024/Meghalaya/213	Field Survey	West Garo Hills	Rongbretgre Landslide 2	Paikan Tura Dalu Road	NH 217	25.396902	90.226635	15-06-2024		1	0	125.69
85	Newspapers & Media/2024/Meghalaya/214	Field Survey	West Garo Hills	Rongbretgre Landslide 3	Paikan Tura Dalu Road	NH 217	25.395992	90.22618	15-06-2024		1	0	125.69
86	Newspapers & Media/2024/Meghalaya/215	Field Survey	West Garo Hills	Rongbretgre Landslide 4	Paikan Tura Dalu Road	NH 217	25.39366	90.225665	15-06-2024		1	0	125.69
87	Newspapers & Media/2024/Meghalaya/216	Field Survey	West Garo Hills	Santogre Landslide 1	Paikan Tura Dalu Road	NH 217	25.355696	90.222252	15-06-2024		1	0	125.69
88	Newspapers & Media/2024/Meghalaya/217	Field Survey	West Garo Hills	Santogre Landslide 2	Paikan Tura Dalu Road	NH 217	25.355738	90.222034	15-06-2024		1	0	125.69
89	Newspapers & Media/2024/Meghalaya/218	Field Survey	West Garo Hills	Kherapara Landslide 1	Paikan Tura Dalu Road	NH 217	25.338717	90.203781	15-06-2024		1	0	125.69
90	Newspapers & Media/2024/Meghalaya/219	Field Survey	West Garo Hills	Kherapara Landslide 2	Paikan Tura Dalu Road	NH 217	25.324709	90.205449	15-06-2024		1	0	125.69
91	Newspapers & Media/2024/Meghalaya/220	Field Survey	West Garo Hills	Kherapara Landslide 3	Paikan Tura Dalu Road	NH 217	25.324395	90.205134	15-06-2024		1	0	125.69

S. No.	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_mm)
92	Newspapers & Media/2024/Meghalaya/221	Field Survey	West Garo Hills	Kherapara Landslide 4	Paikan Tura Dalu Road	NH 217	25.324305	90.205068	15-06-2024		1	0	125.69
93	Newspapers & Media/2024/Meghalaya/222	Field Survey	West Garo Hills	Dopogre Landslide 1	Paikan Tura Dalu Road	NH 217	25.314873	90.200065	15-06-2024		1	0	125.69
94	Newspapers & Media/2024/Meghalaya/223	Field Survey	West Garo Hills	Dopogre Landslide 2	Paikan Tura Dalu Road	NH 217	25.314952	90.200072	15-06-2024		1	0	125.69

Annexure-I (V): Mizoram

S. No.	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_mm)
1	Newspapers & Media/2024/Mizoram/13	Newspapers & Media	Aizawl	Hlimen Landslide	Situated between Meltham and Hlimen in the southern part of Aizawl city, Mizoram		23.68059	92.71768	28-05-2024	6:00 AM	2	28	660.93
2	Newspapers & Media/2024/Mizoram/14	Newspapers & Media	Aizawl	Hunthar Landslide	Western outskirts of Aizawl	NH-54	23.74506	92.71425	28-05-2024		3	0	596.485
3	Newspapers & Media/2024/Mizoram/15	Newspapers & Media	Aizawl	Salem Landslide	Near Aizawl City		23.71769	92.7266	28-05-2024		2	0	596.485
4	Newspapers & Media/2024/Mizoram/27	Newspapers & Media	Aizawl	Falkawn Landslide	Falkawn Village		23.62224	92.72635	28-05-2024		2	2	656.676
5	Newspapers & Media/2024/Mizoram/28	Newspapers & Media	Aizawl	Lungsei Landslide	Lungsei Village		23.45122	92.68159	28-05-2024		2	1	594.249
6	Newspapers & Media/2024/Mizoram/29	Newspapers & Media	Aizawl	Aibawk Landslide	Aibawk Village		23.55912	92.70656	28-05-2024		2	1	536.282
7	Newspapers & Media/2024/Mizoram/49	Newspapers & Media	Mamit	Khantlang Landslide	Khantlang Village in Mamitdistrict		23.75095	92.27094	11-06-2024		2	0	70.7
8	Newspapers & Media/2024/Mizoram/86	Newspapers & Media	Aizawl	Zuangtui Landslide	Aizawl's Zuangtui locality		23.75524	92.73935	02-07-2024	7:30 AM	2	3	

S. No.	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_mm)
9	Newspapers & Media/2024/Mizoram/94	Newspapers & Media	Aizawl	Bawngkawn Landslide	Aizawl's Bawngkawn locality		23.75565	92.72946	02-07-2024	Morning	2	0	338.11
10	Newspapers & Media/2024/Mizoram/95	Newspapers & Media	Aizawl	Zemabawk Landslide	Aizawl's Zemabawk locality		23.73397	92.7463	02-07-2024	Morning	2	3	172.74
11	Newspapers & Media/2024/Mizoram/101	Newspapers & Media	Aizawl	Durtlang Landslide	Durtlang Road		23.76229	92.73362	02-07-2024		3	0	173.43
12	Newspapers & Media/2024/Mizoram/170	Newspapers & Media	Kolasib	Kawnpui Landslide	Near Kawnpui Railway Station		23.99435	92.64393	28-08-2024		2	0	45.79

Annexure-I (VI): Nagaland

S. No.	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_mm)
1	Newspapers & Media/2024/Nagaland/25	Newspapers & Media	Phek	Rekizu Landslide	Rekizu Ward		25.582963	94.298515	28-05-2024		2	1	194.3
2	Newspapers & Media/2024/Nagaland/26	Newspapers & Media	Phek	Kikruma Landslide	Kikruma Village		25.579584	94.220527	28-05-2024		2	0	194.3
3	Newspapers & Media/2024/Nagaland/56	Newspapers & Media	Kiphire	Medical Ward Landslide	Medical ward, Kiphire		25.881651	94.780119	16-06-2024	11:00 AM	2	0	0.21
4	Newspapers & Media/2024/Nagaland/70	Newspapers & Media	Mokokchung	Tongdentsuyong Landslide	Tongdentsuyong ward near butcher shop	NH 2	26.328052	94.524576	28-06-2024	1:30 AM	2	0	100.43
5	Newspapers & Media/2024/Nagaland/77	Newspapers & Media	Zunheboto	DC Hill Landslide	DC Hill West Area		26.012021	94.52143	29-06-2024		2	0	15.989
6	Newspapers & Media/2024/Nagaland/99	Newspapers & Media	Tuensang	Tuensang Landslide	TuensangMokokchung Road	NH 202	26.223056	94.800348	02-07-2024		2	0	122.05
7	Newspapers & Media/2024/Nagaland/102	Newspapers & Media	Kiphire	Shishimi Landslide	Shishimi Village Area		25.816009	94.616317	02-07-2024		3	0	301.5

S. No.	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_mm)
8	Newspapers & Media/2024/Nagaland/103	Newspapers & Media	Zunheboto	DIET, Zunheboto Landslide	South point east colony		26.006772	94.521634	01-07-2024		2	0	32.2
9	Newspapers & Media/2024/Nagaland/104	Newspapers & Media	Tuensang	K.Wongthu Landslide	Kuthur Village		26.185832	94.841673	03-07-2024		3	0	17.91
10	Newspapers & Media/2024/Nagaland/116	Newspapers & Media	Shamator	Chessore Landslide	Keku colony of Chessore area		26.090935	94.746775	05-07-2024		3	0	106.7
11	Newspapers & Media/2024/Nagaland/124	Newspapers & Media	Kohima	Seca Khel Landslide	Kigwema-Mima-Chakhabama Road		25.607096	94.12725	12-07-2024		3	0	18.59
12	Newspapers & Media/2024/Nagaland/128	Newspapers & Media	Zunheboto	Alahuto Landslide	Alahuto Colony of Zunheboto Town		26.015899	94.531077	07-07-2024	4:00 PM	2	0	10.85
13	Newspapers & Media/2024/Nagaland/132	Newspapers & Media	Wokha	Wokha Landslide	Near Wokha village 'B' Khel	NH 2	26.107402	94.255575	15-07-2024	5:00 AM	2	0	3.34
14	Newspapers & Media/2024/Nagaland/135	Newspapers & Media	Wokha	Tchuyiro colony Landslide	Tchuyiro colony area in Wokha Town		26.09928	94.25861	22-07-2024	2:00 AM	3	0	15.09
15	Newspapers & Media/2024/Nagaland/136	Newspapers & Media	Wokha	Lakhuti Landslide	Near football ground at Lakhuti Village		26.2956	94.246652	22-07-2024	1:30 AM	2	0	34.58
16	Newspapers & Media/2024/Nagaland/138	Newspapers & Media	Kohima	Pezielietsie Colony Landslide	Pezielietsie Colony in Ward-3 North Block, Kohima		25.680683	94.103026	26-07-2024		2	0	1.29
17	Newspapers & Media/2024/Nagaland/149	Newspapers & Media	Kohima	Dzüü Landslide	Just before Dzüü bridge		25.671317	94.18373	04-08-2024		2	0	0.4
18	Newspapers & Media/2024/Nagaland/156	Newspapers & Media	Kiphire	Lophukhong Landslide	Near Lophukhong Village		25.699199	94.930126	12-08-2024		3	0	57.46
19	Newspapers & Media/2024/Nagaland/158	Newspapers & Media	Kohima	Zubza Landslide	Kohima Dimapur Highway	NH 29	25.724445	94.034266	14-08-2024		3	0	4.66
20	Newspapers & Media/2024/Nagaland/168	Newspapers & Media	Kohima	Khonoma Landslide	Between Jotsoma and Khonoma		25.670326	94.04216	23-08-2024	Morning	3	0	45.29

S. No.	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_mm)
21	Newspapers & Media/2024/Nagaland/169	Newspapers & Media	Wokha	Etsuchuka Colony Landslide	Etsuchuka Colony, Wokha		26.098734	94.254992	26-08-2024	3:00 PM	2	0	3.6
22	Newspapers & Media/2024/Nagaland/171	Newspapers & Media	Wokha	Longsa Landslide	Between Longsa Junction and Dikhu Bridge	NH 202	26.249363	94.537726	29-08-2024		2	0	21.63
23	Newspapers & Media/2024/Nagaland/173	Newspapers & Media	Mokokchung	Mokokchung Landslide	Between MokokchungTu engsang Road		26.302047	94.586325	26-08-2024		3	0	20.23
24	Newspapers & Media/2024/Nagaland/174	Newspapers & Media	Chumoukedima	Pherima Landslide	Between Pherima and Paglapahar	NH 29	25.773953	93.913722	03-09-2024	11:00 PM	3	6	33.41
25	Newspapers & Media/2024/Nagaland/177	Newspapers & Media	Kohima	KMC Dumping Landslide	Near old Kohima Municipal Council (KMC) dumping site	NH 29	25.710966	94.068528	09-09-2024	5:00 PM	3	0	45.03
26	Newspapers & Media/2024/Nagaland/183	Newspapers & Media	Kohima	KMC Dumping Landslide	Near old Kohima Municipal Council (KMC) dumping site	NH 29	25.710898	94.068536	04-10-2024		3	0	16.38

Annexure-I (VII): Sikkim

S. No	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_mm)
1	Newspapers & Media/2024/Sikkim/9	Newspapers & Media	Pakyong	Pakyong DC office Landslide	Near under construction DC office		27.24534	88.61161	17-05-2024	Afternoon	2	0	1.803
2	Newspapers & Media/2024/Sikkim/33	Newspapers & Media	Gangtok	Matim Landslide	Maltin and BOP areas		27.62714	88.67311	01-06-2024		3	0	0.54

S. No	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_mm)
3	Newspapers & Media/2024/Sikkim/34	Newspapers & Media	Gangtok	Tashi Landslide	Gangtok Mangan Road		27.37194	88.61639	01-06-2024		3	0	0
4	Newspapers & Media/2024/Sikkim/43	Newspapers & Media	Namchi	Majua Landslide	Yangang area		27.29403	88.41938	10-06-2024	6:30 AM	2	3	2.4
5	Newspapers & Media/2024/Sikkim/45	Newspapers & Media	Mangan	Mangan Landslide	Mangan Area		27.50028	88.53445	13-06-2024		3	6	99
6	Newspapers & Media/2024/Sikkim/46	Newspapers & Media	Gangtok	Shantinagar Landslide	Shantinagar Area in Singtam	NH 10	27.23553	88.50579	11-06-2024		3	0	10
7	Newspapers & Media/2024/Sikkim/47	Newspapers & Media	Gangtok	Pani House Landslide	Pani House area in Gangtok		27.32167	88.60993	11-06-2024		3	0	13.6
8	Newspapers & Media/2024/Sikkim/48	Newspapers & Media	Mangan	Zeema Landslide	Zeema region in North Sikkim		27.86372	88.54756	11-06-2024		3	0	13.6
9	Newspapers & Media/2024/Sikkim/50	Newspapers & Media	Mangan	Pakshep Landslide	Pakshep area in Mangan		27.50669	88.55658	13-06-2024		2	3	99
10	Newspapers & Media/2024/Sikkim/51	Newspapers & Media	Mangan	Sangkalang Landslide	Near Sangkalang Bridge		27.5086	88.52691	13-06-2024		3	0	99
11	Newspapers & Media/2024/Sikkim/74	Newspapers & Media	Gangtok	Lingding Landslide	LingdingSichey Road		27.32468	88.60395	28-06-2024		3	0	11.56
12	Newspapers & Media/2024/Sikkim/123	Newspapers & Media	Gangtok	Makha Landslide	Near Sisney, close to 10th Mile in Makha		27.35164	88.32418	12-07-2024		2	1	3.81
13	Newspapers & Media/2024/Sikkim/133	Newspapers & Media	Mangan	Munsi Thang Landslide	Enroute Lachen		27.82149	88.5491	21-07-2024		2	0	0.46
14	Newspapers & Media/2024/Sikkim/137	Newspapers & Media	Mangan	Toong Landslide	Between Naga and Rel Ward areas		27.56807	88.65081	22-07-2024		3	0	0.38
15	Newspapers & Media/2024/Sikkim/162	Newspapers & Media	Gangtok	Balutar Landslide	NHPC's 510 MW Teesta Stage 5 dam in Balutar, Sikkim		27.25169	88.45966	20-08-2024	Morning	2	0	81.15
16	Newspapers & Media/2024/Sikkim/172	Newspapers & Media	Pakyong	Pakyong Landslide	Near Pakyong Airport		27.23968	88.59367	29-08-2024		2	0	1.64

S. No.	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_mm)
17	Newspapers & Media/2024/Sikkim/179	Newspapers & Media	Mangan	Rang Rang Landslide	Near Rang Rang Landslide		27.47775	88.52716	30-09-2024		3	0	5.929
18	Newspapers & Media/2024/Sikkim/180	Newspapers & Media	Gangtok	Lower Tintek Landslide	Marchak, Lower Tintek		27.37807	88.49725	30-09-2024		3	0	0.569
19	Newspapers & Media/2024/Sikkim/181	Newspapers & Media	Soreng	RibdiBhareng Landslide	Between Ribdi and Bhareng		27.17605	88.09037	30-09-2024		2	0	0

Annexure-I (VIII): Tripura

S. No.	Slide ID	Source	District	Landslide Name	Location	NH/SH Affected	Latitude	Longitude	Date of Event	Time of Event	Type of Event	Casualty	Rainfall GPM (24hr_mm)
1	Newspapers & Media/2024/Tripura/54	Newspapers & Media	West Tripura	Mungiakami Landslide	Mile 35 in Mungiakami region		23.878866	91.705846	15-06-2024		3	0	0.53
2	Newspapers & Media/2024/Tripura/163	Newspapers & Media	South Tripura	Debipur Landslide	Debipur region		23.167876	91.516866	19-08-2024		2	3	42.19
3	Newspapers & Media/2024/Tripura/164	Newspapers & Media	Khowai	Teliamura Landslide	Teliamura region		23.839927	91.634953	19-08-2024		2	1	46.35
4	Newspapers & Media/2024/Tripura/165	Newspapers & Media	Dhalai	Saikar Landslide	Near Saikar Bazar		24.109649	91.908101	19-08-2024		3	0	48
5	Newspapers & Media/2024/Tripura/166	Newspapers & Media	South Tripura	Gardang Landslide	Gardang Panchayat near Shanti bazar		23.312969	91.524158	22-08-2024	Early Morning	2	7	677
6	Newspapers & Media/2024/Tripura/167	Newspapers & Media	South Tripura	Nutan Bazar Landslide	Near Nutan bazar water treatment plant		23.423169	91.755621	22-08-2024		2	0	484.05

* 914 landslides reported by Tripura SDMA can be found in Tripura SDMA Report.

ANNEXURE-II: LANDSLIDE VERIFICATION ON SATELLITE IMAGERY AND
GOOGLE EARTH IMAGERY

Newspapers & Media/2024/Sikkim/162



Newspapers & Media/2024/Sikkim/179

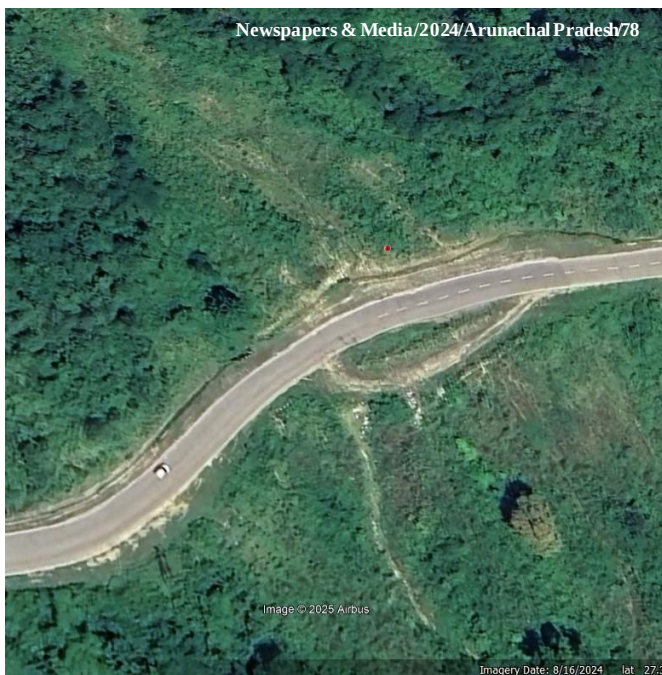
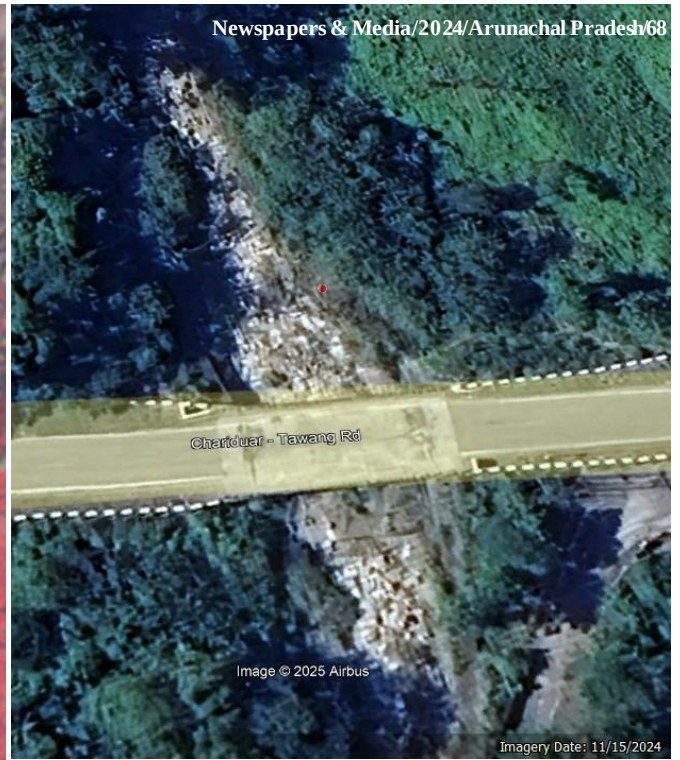
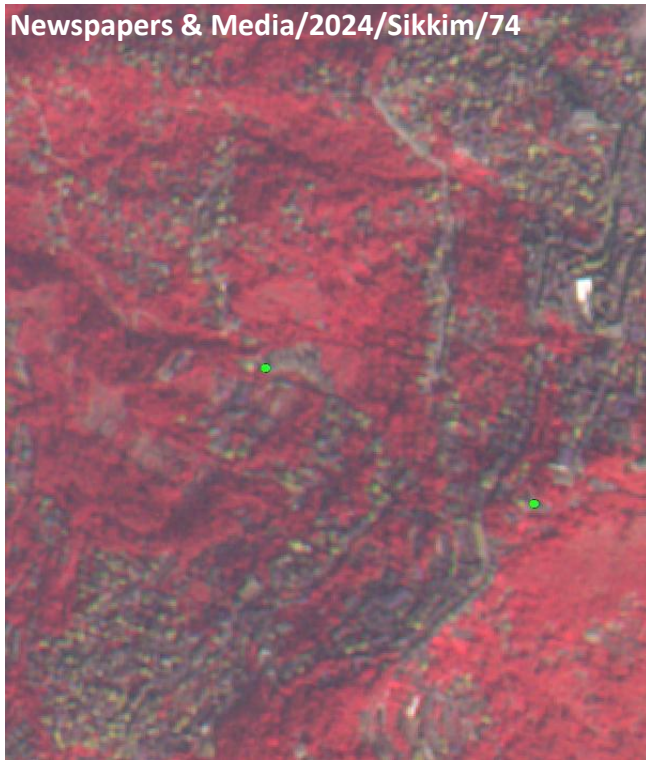


Newspapers & Media/2024/Sikkim/180



Newspapers & Media/2024/Sikkim/137





ANNEXURE-III: NEWSPAPER (CLIPPINGS & DIGITAL ARCHIVE)/ONLINE MEDIA

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THE TIMES OF INDIA

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THIS STORY IS FROM APRIL 25, 2024

Massive landslide in Arunachal Pradesh's Dibang along China border, road connectivity cut off

TOI City Desk / TIMESOFINDIA.COM / Apr 25, 2024, 11:33 IST

CM Pema Khandu of Arunachal Pradesh addresses the disruption caused by a landslide on National Highway-33 in Dibang Valley near China. He assures quick restoration of connectivity between Hunli and Anini towns.

ITANAGAR: A significant landslide has severed road access in Arunachal Pradesh's Dibang Valley, a region bordering China. The incident occurred on

Delhi 16°C

Hindustan Times

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Meghalaya landslide: Four bodies recovered

By David Laitphiang

Jun 01, 2024 05:41 PM IST

Torrential rains and fierce winds triggered a landslide in a remote village in Meghalaya's Sohra, burying two houses and claiming four lives



Officials said the teams had to carry out the rescue operations with only small tools and implements as no heavy machinery could be brought since the village is not connected by road. (Representative Image)

The Assam Tribune

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Sikkim: Landslide in Majua, Yangang claims 3 lives; destroys seven houses

The Shillong Times

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Tripura: 3 of family among 7 killed in landslides due to heavy rainfall

By Agnes Date: August 20, 2024



DH

Home India Mizoram

Three feared dead as rain-triggered landslide buries building in Mizoram

On Sunday, the state disaster management and rehabilitation department had issued an alert in anticipation of heavy rainfall and storms.

PTI

Last Updated: 02 July 2024, 01:04:07



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Fatal Mine Collapse In Mizoram: Landslide Kills 15 Workers During Heavy Rain

Mizoram experienced a severe landslide today in Aizawl city due to heavy rains, leading to the collapse of a stone mine. The disaster resulted in the deaths of 15 workers, with many more still buried under the debris.

Edited By: simran rajpal Updated: May 28, 2024 13:19 IST



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Arunachal

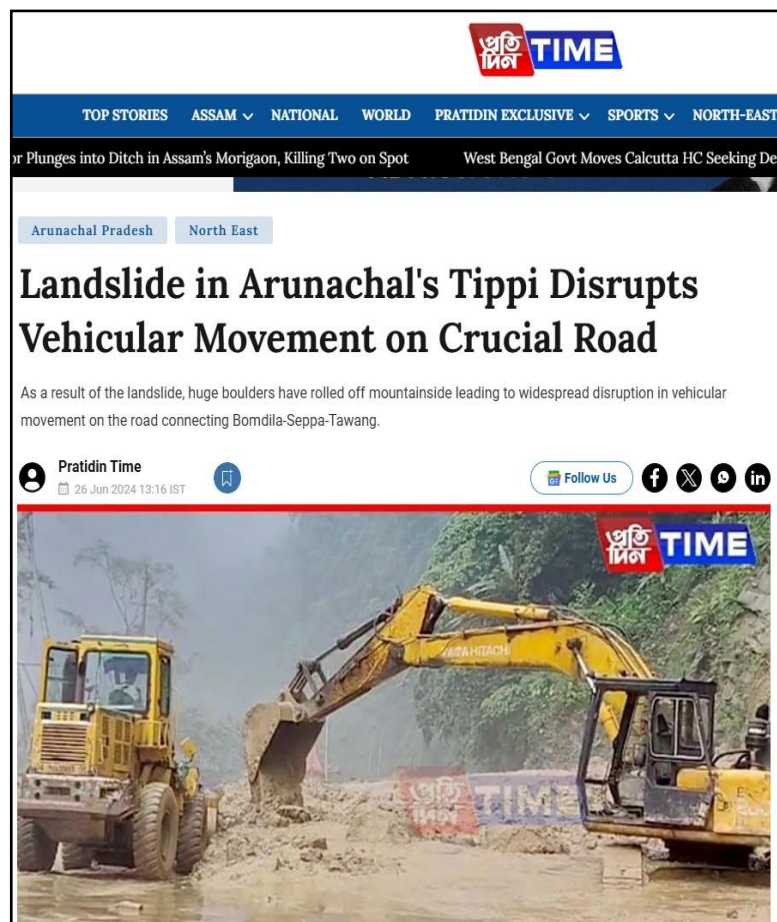
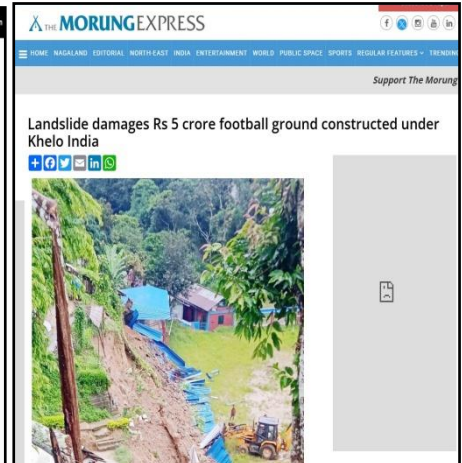
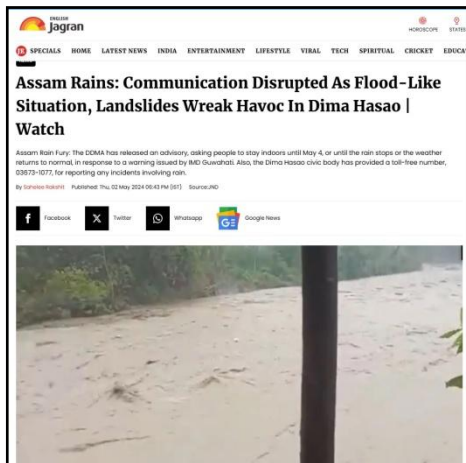
Arunachal: Heavy Landslide on Margherita-Changlang road

The District administration issued an orders to prohibit all kinds of vehicle movement on Margherita-Changlang road till restoration work is completed.

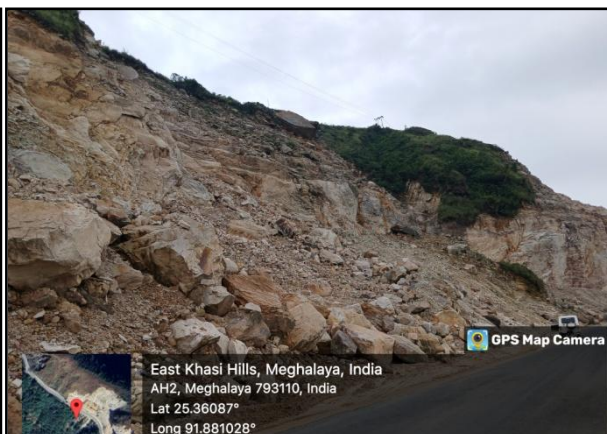
Last Updated: June 16, 2022

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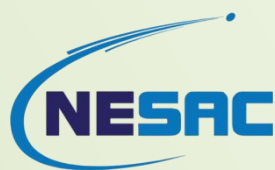
ANNEXURE-IV: FIELD SURVEY POTOGRAPHS





ANNEXURE-V: INVENTORY ATTRIBUTES

Slide ID	Unique Identifier for Landslide
Source	Newspaper/Media, SDMA, GSI or From Field
Slide No	Serial number
State	State in NER where landslide occurred
District	District where landslide occurred
Name	Name of the event
Location	Specific location from SDMA reports. General location – neighbourhood, village name etc from newspapers
NH/SH Cutoff/Affected	If National Highway, State Highway or any other important road was cutoff or affected
Latitude	Exact Latitude from GSI/SDMA reports General Latitude from Newspaper Articles
Longitude	Exact Longitude from GSI/SDMA reports General Longitude from Newspaper Articles
Date of Event	Date of the event
Time of Event	Time of event wherever possible
Type of Event	Type 1, 2 or 3 (explained in the next section)
Casualty	Casualty figures
Rainfall (24 hrs)	24 hours rainfall prior to the event



North Eastern Regional Node for Disaster Risk Reduction
North Eastern Space Applications Centre
Dept. of Space, Govt. of India
Umiam-793103, Meghalaya